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Direct feature extraction and image co-registration of morphological structure from Bayer pattern raw planetary images

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ABSTRACT

Feature extraction from planetary remote sensing images is a primary image processing task for object recognition, crater counter, morphological structure dimension and unpaired image co-registration. The paper presents a novel methodology to extract features directly from Bayer Pattern raw planetary images. Gradient information is extracted from Bayer Pattern image using standard edge operator that follows Color Difference Constancy (CDC) assumption. The proposed method's advantage is that it can skip the computationally intensive image processing pipeline and can have the Bayer Pattern raw planetary image flow directly for useful information extraction. Sobel Edge Detector is applied on Indian Mars Color Camera (MCC) Bayer Pattern raw images, and Gradient Magnitude Map (GMM) is generated at different Martian terrains. In addition, we have developed a direct image co-registration approach for MCC Bayer intensity raw image with respect to Mars Digital Image Model (MDIM) 2.1 reference using Mode-Mean Combo Patch Filler and Gradient Intensity induced Scale Invariant Feature Transform (GI-SIFT) based feature matching. The outlier matched points are removed by Feature Similarity Score guided Random Sample Consensus (FSS-RANSAC) estimation technique. The visual evaluation and quantitative metrics indicate that GMM from MCC Bayer Pattern raw image has negligible degradation with respect to GMM extracted from MCC demosaic image. The Root Mean Square Error (RMSE) is computed at different Mars regions, and it is found that the average image co-registration accuracy is less than 0.5 pixel.

1. Introduction

Feature extraction from remote sensing images provides superlative information for subsequent stages of image analysis (Nevatia & Babu, 1980). The computer vision algorithms are exploited to extract crucial information from multispectral satellite images which generally have at least three spectral channels. To generate a remotely sensed natural color image of a specific area over a planet, an orbital camera generates the pure color samples at each image pixel from Red (R), Green (G) and Blue (B) wavelength channels. The three-sensor camera can produce high quality color image of the planet for different space-borne applications; but the instrument is large in size, and the weight of the camera system is also at the higher margin. Generally, for inter-planetary missions, weight is a major constraint for design of a payload or instrument. This requirement forces to explore an alternative solution such as mono-sensor camera, which is popular in today's digital world and specifically embedded in smartphones today. In a mono-sensor camera, images are captured with only one sensor covered by a Color Filter Array (CFA) that forms Bayer Pattern mosaic schema (Bayer, 1976). In Bayer Pattern image, only one sample out of three color components is captured by each pixel element. The single channel image is converted to a color image using demosaicking process. The image demosaicking (Li, Gunturk & Zhang, 2008) is a digital image process to generate color image from Bayer Pattern CFA image by interpolating missing color components at each image pixel.

Mars Color Camera (MCC) (Arya et al., 2015) on-board Indian Mars Orbiter Mission (MOM) (Arunan & Satish, 2015) spacecraft is also a Bayer Pattern camera that captures planet Mars surface at varying spatial resolution due to highly elliptical orbit of MOM. MCC operates in visible region of electromagnetic spectrum, and generates RGB Bayer Pattern structure from a frame detector array of 2048*2048 elements on a pixel pitch of 5.5μ. MCC is the eye of MOM that captures full disc image

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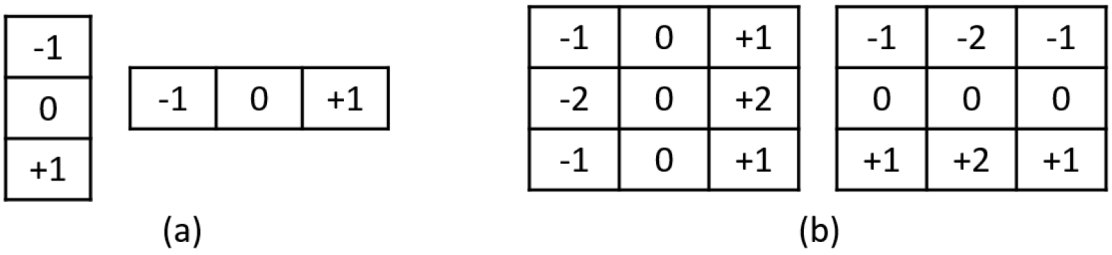

Fig. 1. Edge Operators (a) Central Difference Operator (b) Sobel Operator.

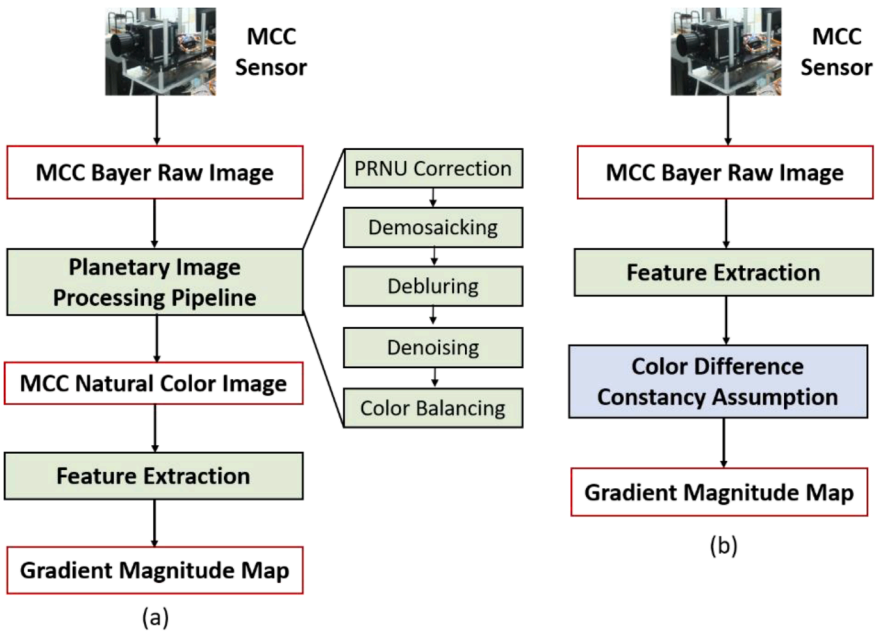

Fig. 2. Feature Extraction Pipelines (a) Conventional Feature Extraction Pipeline (b) Proposed Direct Feature Extraction Pipeline for MCC Bayer Pattern Raw Image.

of Mars at a spatial resolution of 4.0 km from apopsis, and can take closer view of an areo-morphological feature target at a relatively higher spatial resolution of 15.0 m from preapsis. MCC Long Term Archive (LTA) is a fully compliant Planetary Data System (PDS) imagery collection that contains all the relevant information with each object, and is available to planetary scientists in public domain (Moorthi et al., 2015). More than 1000 unique MCC images over Mars contain a rich treasure of information to understand the latest Martian surface phenomenon that includes dust devil monitoring, changes in Martian ice cap and new impact crater formation (Chevrier & Mathe, 2007).

The paper highlights a novel methodology for direct feature extraction from MCC Bayer Pattern raw image. MCC raw data received on ground needs to be processed through a series of radiometric and geometric image processing steps in order to generate a MCC color image. The major data processing step that consumes significant amount of time involves demosaicking and denoising of MCC images. The image processing pipeline developed is complex and computationally intensive (Misra et al., 2019). We have attempted to extract gradient feature directly from MCC Bayer Pattern raw image using edge detectors that follow Color Difference Constancy (CDC) assumption (Zhou et al., 2021). Central Difference Edge Operator and Sobel Edge Detector (Kanopoulos, Vasanthavada & Baker, 1988) are primarily applied for direct gradient extraction from MCC Bayer Pattern raw image. We have developed a procedure for image co-registration from MCC raw image using Mode-Mean Combo Patch Filler and Gradient Intensity induced Scale Invariant Feature Transform (GI-SIFT) (Lowe, 1999) based feature matching along with Feature Similarity Score guided Random Sample Consensus (FSS-RANSAC) (Fischler & Bolles, 1981) outlier removal technique. The experimental evaluation indicates that the Gradient Magnitude Map (GMM) extracted from MCC Bayer Pattern raw image and MCC demosaic image produces the almost same circumference trace for crater boundary, Martian mons and valleys over Mars surface. The GMM results generated are quantitatively evaluated with respect to Mars Digital Image Model (MDIM) 2.1 (Archinal et al., 2004) reference to demonstrate the robustness of our proposed method. The image co-registration exercises are also performed at different regions of Mars terrains, and it is observed that the proposed direct image co-registration approach achieve sub-pixel level relative geometric accuracy. The comparison shows that co-registration estimates from direct MCC Bayer Pattern raw image are also in close proximity with MCC demosaic image. The major contributions of our work can be summarized as follows.

1) Developed a novel procedure to extract essential features directly from MCC Bayer Pattern raw planetary image using edge detector that holds CDC assumption.
2) Applied direct feature extraction method on raw planetary remote sensing images over Mars and deduce pivotal information for the next level of image processing operations.
3) Designed Mode-Mean Combo Patch Filler to generate intensity image directly from MCC Bayer Pattern raw image, and proposed GI-SIFT feature matching with FSS-RANSAC based outlier removal technique for sub-pixel image co-registration.
4) Direct feature extraction and co-registration performance is assessed visually at varying morphological structures. Quantitative evaluation shows close agreement with respect to results achieved from

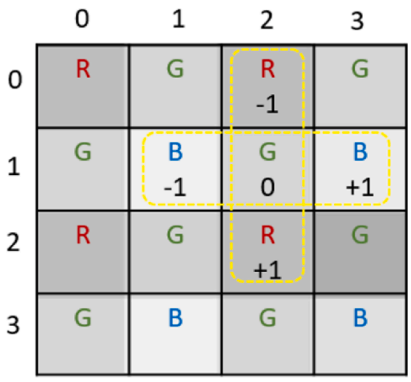

Fig. 3. Gradient Magnitude Extraction from MCC Bayer Pattern Raw Image.

MCC demosaic image. The proposed image registration method outperforms state-of-the-art feature-based techniques.

The remainder of this paper is organized as follows. Section-II describes the direct feature extraction methodology and the image co-registration approach. Section-III analyses the visual and quantitative performance of GMM and the RMSE achieved for image co-registration with comparison with state-of-the-art techniques. Section-IV presents the concluding remarks.

2. Methodology developed

2.1. Feature extraction from MCC Bayer Pattern raw image

The MCC Bayer Pattern raw image is used directly to extract gradient intensity map information. The gradients are computed based on derivatives of image pixel with respect to the horizontal direction component x and the vertical direction component y. The gradient operator applied on MCC Bayer Pattern raw image should satisfy the following criteria.

1) Near to zero gradient intensity value at homogeneous area.
2) High difference estimate on the sloping surfaces.
3) Non-zero measurement at the beginning or end of a ramp.

The above conditions are met if an edge detector has alternate zero and non-zero coefficients, such that it can ensure the Color Difference Constancy (CDC) assumption. The demosaicking algorithms heavily presume CDC to generate full color image due to superior Peak Signal-to-Noise Ratio (PSNR) performance. We have extracted gradient information from MCC Bayer image using gradient operators that satisfy CDC assumption. The Central Difference Operator and Sobel Edge Filter, as shown in Fig. 1, are selected to extract prominent gradient feature from MCC Bayer Pattern raw image.

The key idea is to apply gradient filter directly on MCC Bayer Pattern

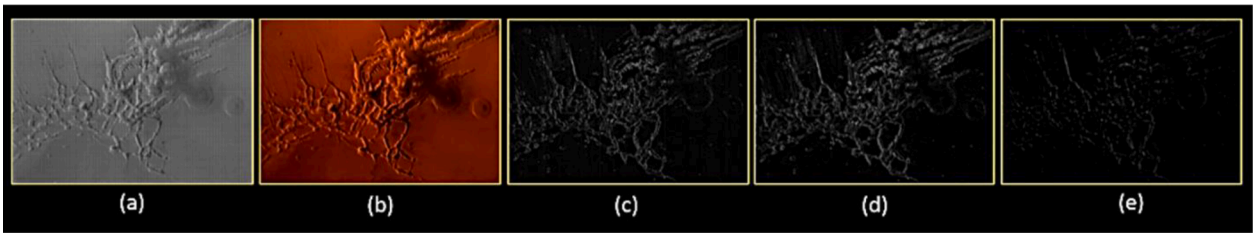

Fig. 4. (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) Gradient Magnitude Map from Bayer (d) Gradient Magnitude Map from Demosaic (e) Gradient Difference Image.

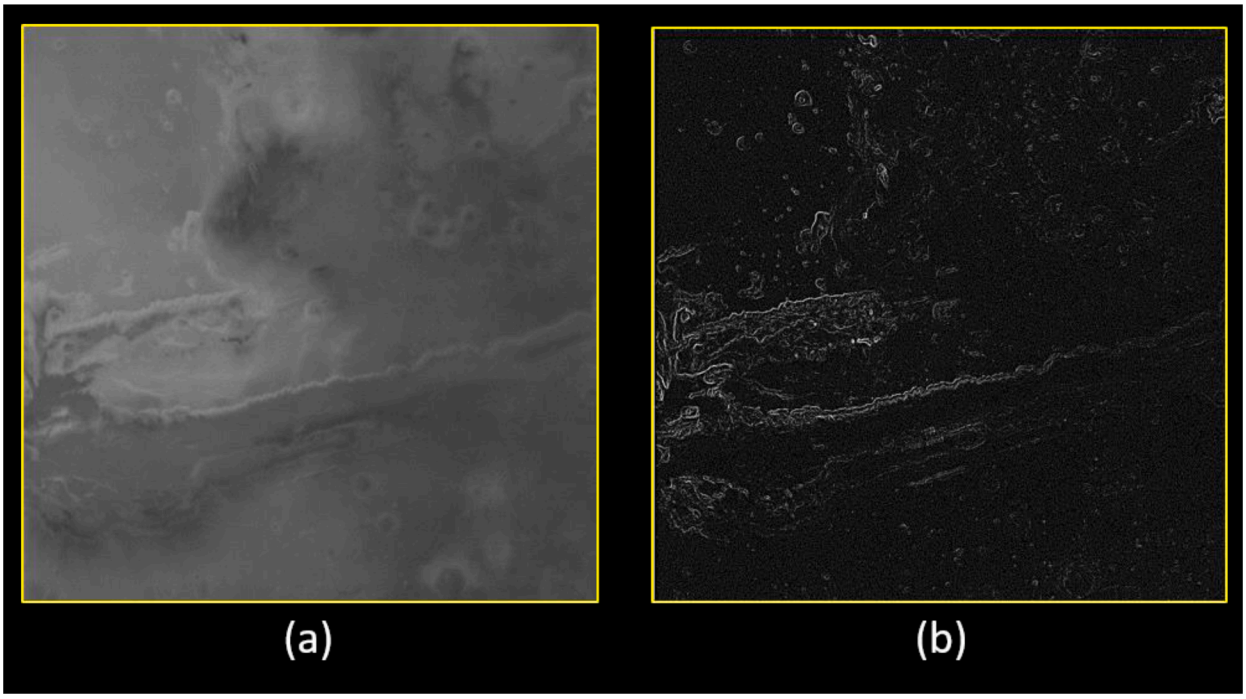

Fig. 5. (a) MCC Bayer Intensity Raw Image (b) Gradient Magnitude Map (GMM) directly from MCC Bayer Intensity Raw Image.

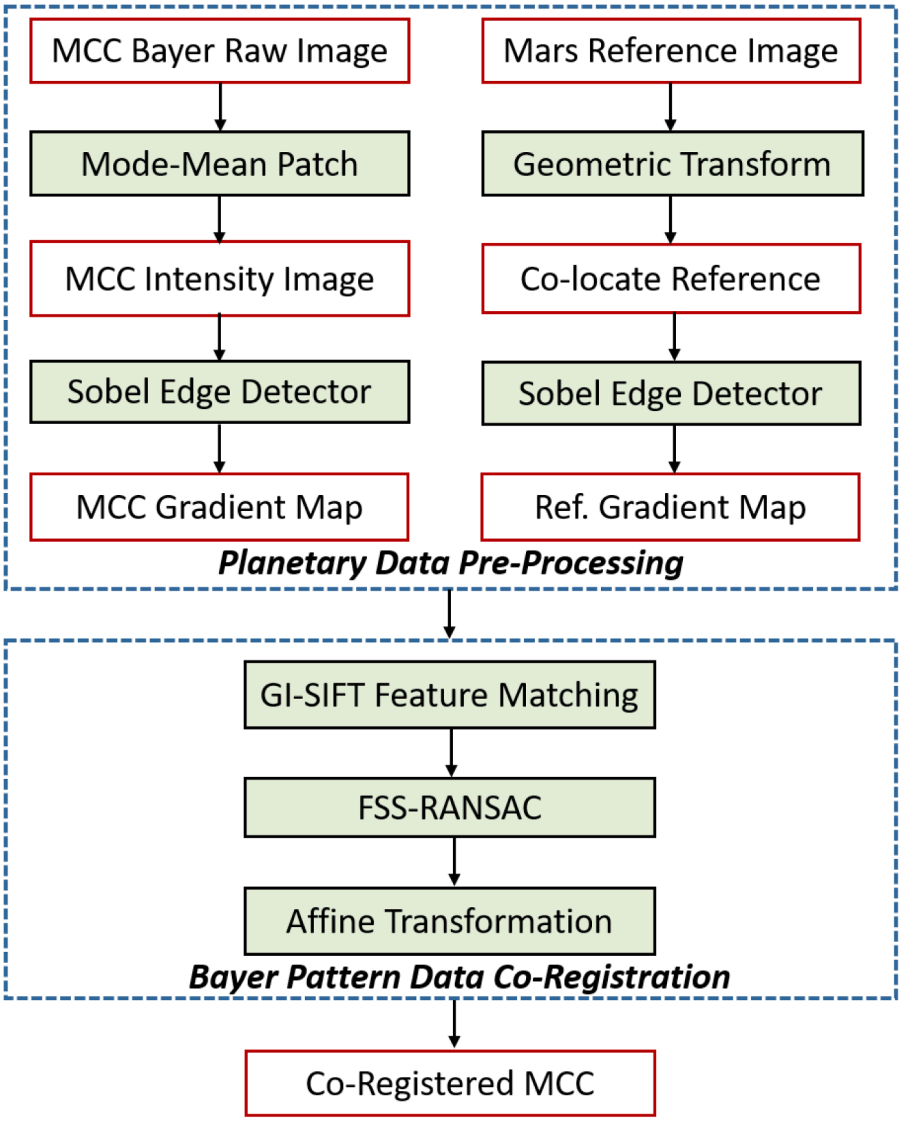

Fig. 6. Direct Bayer Pattern Raw Image Co-Registration Workflow.

Table 1
MCC Bayer Pattern Raw Image Details.

PDS File Name	Imaging Time (YYYY-MM-DD, HH:MM:SS)	Spatial Resolution (in m)	Region Covered
MCC_MEB_20141227T020400953_D_D32.IMG	2014–12-27, 02:04:00	453	Arisa Mons
MCC_MEB_20150117T213420018_D_D32.IMG	2015–01-17, 21:34:20	468	Gale Crater
MCC_MEB_20150423T072306117_D_D32.IMG	2015–04-23, 07:23:06.117	44	Pital Crater
MCC_MEB_20150903T221933652_D_D32.IMG	2015–09-03, 22:19:33.652	292	Tharsis Tholus

Table 2
Experimental Parameters Setting for the Proposed Method.

S.No.	Experimental Parameters	Specified Value
1.	Sigma	4.0
2.	Window Size	5.0
3.	Pre-defined Threshold	0.75
4.	FSS Threshold	0.8

raw image. Both the edge operators in Fig. 1 follow the CDC, which is essential for gradient intensity map generation from raw MCC data. The central difference gradient operator can be defined as:

$$D_x(x,y) = I(x+1,y) - I(x-1,y) \quad (1)$$

$$D_y(x,y) = I(x,y+1) - I(x,y-1) \quad (2)$$

where D_x and D_y are the image gradients in horizontal and vertical direction respectively,

$I(x,y)$ is the intensity value at pixel location (x,y) of MCC Bayer pattern image.

Instead of performing end-to-end radiometric corrections and demosaicking of MCC images, we convolve the filter template shown in Fig. 1 (a) with MCC Bayer Pattern image, which satisfies the conditions stated above and directly extracts the gradient information. Fig. 2 shows

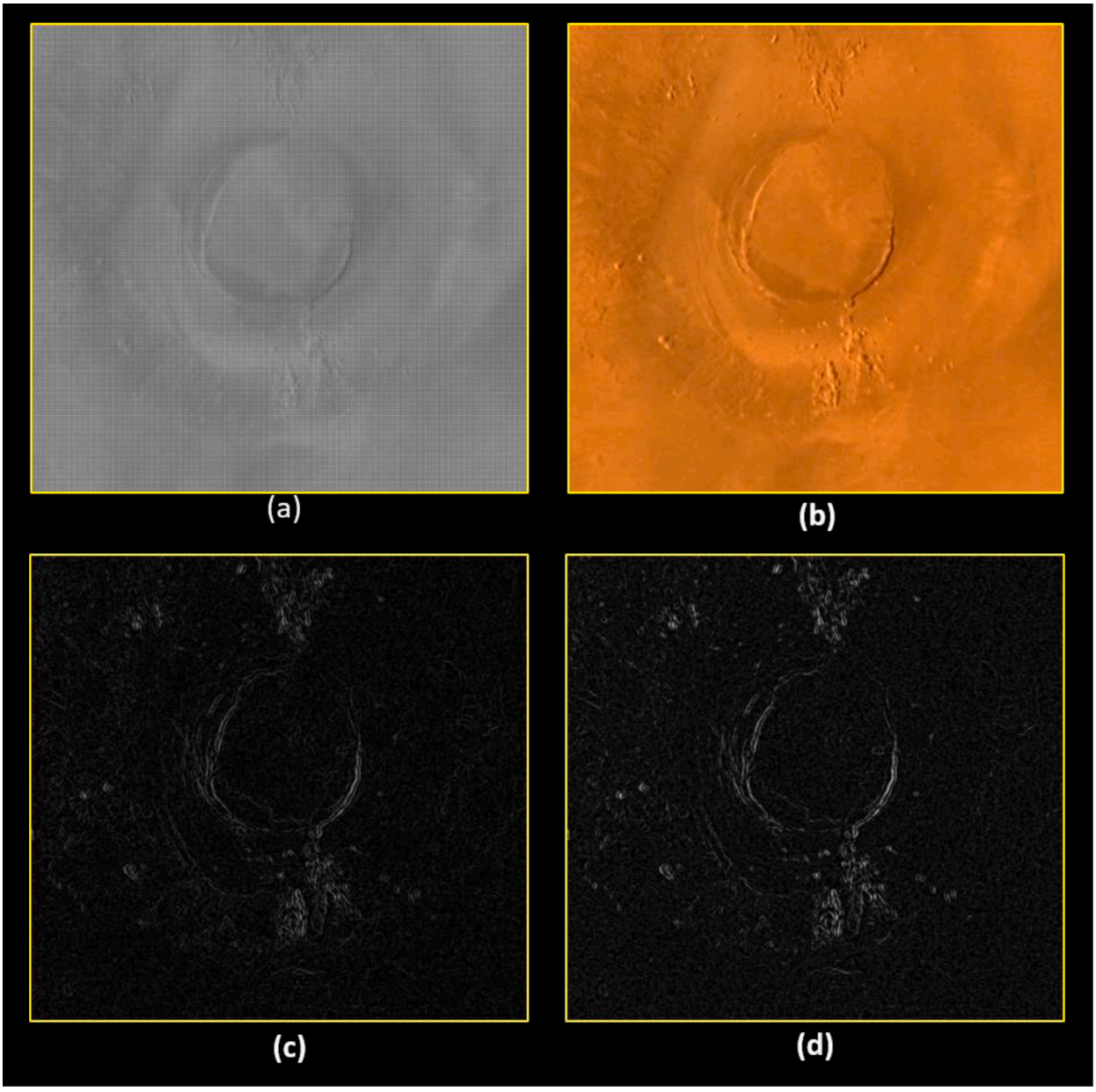

Fig. 7. Direct Feature Extraction at Arisa Mons (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

the conventional and the proposed feature extraction pipelines. The planetary remote sensing image processing pipeline is computationally intensive in conventional mode due to series of image correction procedures. Photo Response Non-Uniformity (PRNU) correction is the first processing step that normalizes the detector response using pre-launch radiometric calibration measures as initial input source (Kinch et al., 2020; Qiao et al., 2021). MCC PRNU corrected Bayer Pattern raw image is reconstructed using demosaicking procedure to generate natural color image of Martian surface (Kiku et al., 2016). It is observed that MCC demosaicked image is little bit blurred due to imaging in a highly elliptical orbit and at uneven solar illumination conditions. Deblurring technique is employed to restore the MCC image, and to enhance the morphological feature details (Dong et al., 2011; Sharif et al., 2023). The enhancement results into noise in homogenous region, and so the image needs to be denoised to suppress the unwanted information (Talebi & Milanfar, 2013; Singh et al., 2021). Denoising using wavelets and deep neural network reduces the noise in planetary remote sensing images (Thukral et al., 2022; Thukral, Kumar & Arora, 2019). Finally, the MCC image is color balanced to adjust the radiance intensities for better Mars landscape representation (Mathew et al., 2017; Yu et al., 2016). The planetary processed images are then used for image level tasks such as segmentation to extract essential features by learning based methods. An adversarial transformer model can be used for 2D medical image segmentation (You et al., 2022a). Contrastive Voxel-wise Representation Learning (CVRL) and Contrastive Distillation are recent popular methods for semi-supervised image segmentation (You et al., 2022b; You et al., 2022c; You et al., 2023a; You et al., 2023b). Incremental-transfer learning and Wasserstein distance-guided approach are used for multi-domain image segmentation (You et al., 2022d; You et al., 2020). The image segmentation in general by learning methods are challenging due to class imbalance, limited labeled data, and domain adaptation (You et al., 2022e; You et al., 2023c). The motivation behind this research work is to skip the heavy weight planetary image processing pipeline for feature extraction. The proposed method extract features directly from raw Bayer pattern planetary images which can be used for diverse planetary science exploration. Our proposed method,

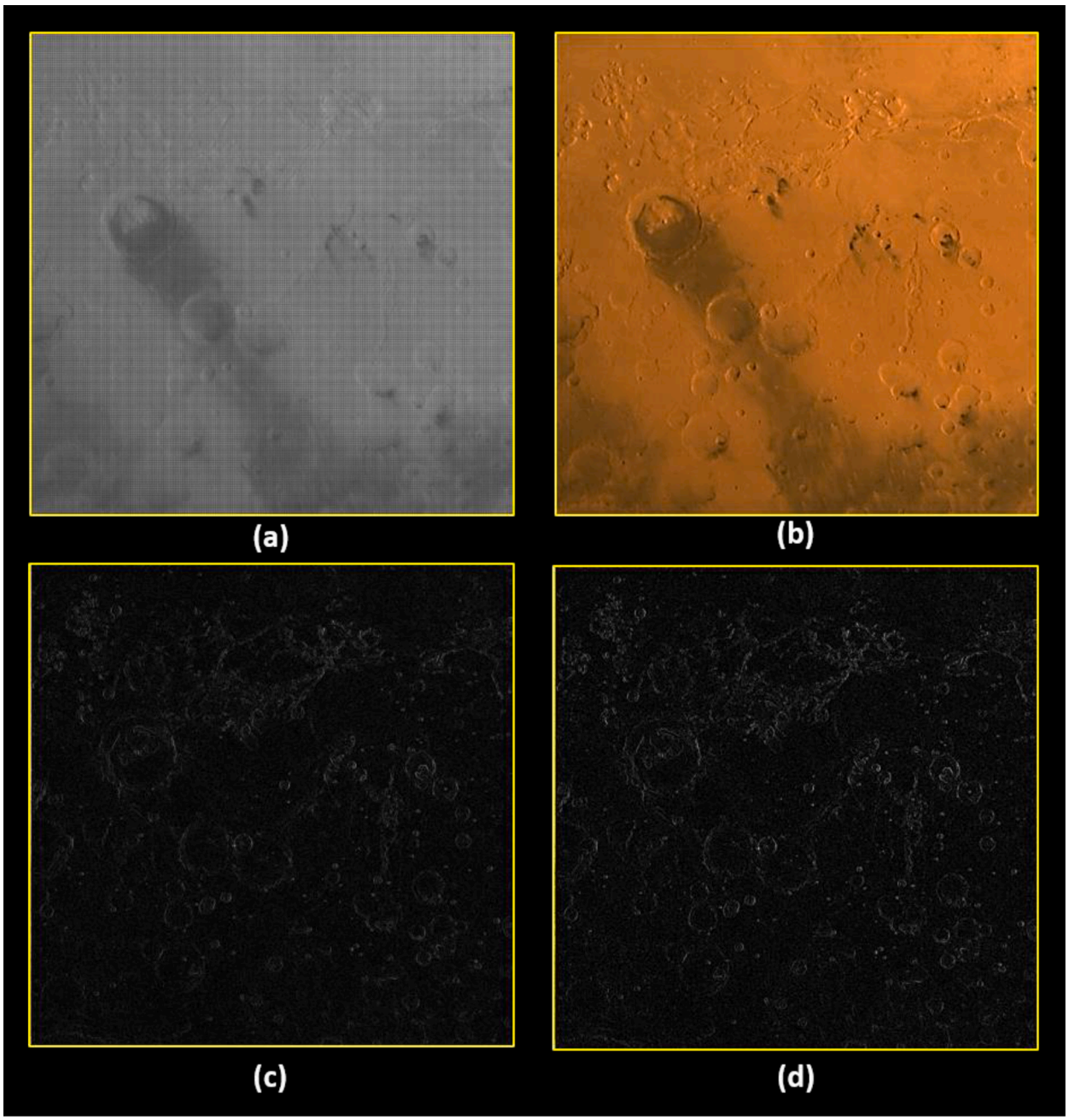

Fig. 8. Direct Feature Extraction around Gale Crater region (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

shown in Fig. 2 (b), skips this heavy load processing sequence, and putative features are extracted directly from MCC Bayer Pattern raw image. The methodology developed relies on a simple design and robust framework for feature extraction and co-registration of raw Bayer pattern planetary remote sensing images. To understand the procedure in detail, an illustrative diagram is shown in Fig. 3, pertaining to a typical MCC RGGB Bayer Pattern scheme and how feature is extracted unambiguously from MCC raw data. The example explains how feature is extracted using CDC compliant gradient operator at MCC pixel location.

The two input MCC pixels for coefficients 1 and −1 in the convolution templates are from the same wavelength channel, so differences are always computed on same spectral channel pixels. The gradient magnitude is computed at location (1, 2) in Fig. 3, and it is formulated as follows.

$$D^{B}_{x(1,2)} = I^{B}_{(1,3)} - I^{B}_{(1,1)} \tag{3}$$

$$D^{R}_{y(1,2)} = I^{R}_{(2,2)} - I^{R}_{(0,2)} \tag{4}$$

where $D^{B} and D^{R}$ are the gradients of blue and red spectral channels, $I^{B}_{(x,y)} and I^{R}_{(x,y)}$ are the intensity values of blue and red spectral channels at pixel location (x, y).

It is observed that if pixel locations (x_1, y_1) and (x_2, y_2) are lie within a close neighborhood, then

$$I^{G}(x_1, y_1) - I^{G}(x_2, y_2) \approx I^{R}(x_1, y_1) - I^{R}(x_2, y_2) \tag{5}$$

$$I^{G}(x_1, y_1) - I^{G}(x_2, y_2) \approx I^{B}(x_1, y_1) - I^{B}(x_2, y_2) \tag{6}$$

Using equation (3) and (4) with equation (5) and (6), we can deduce

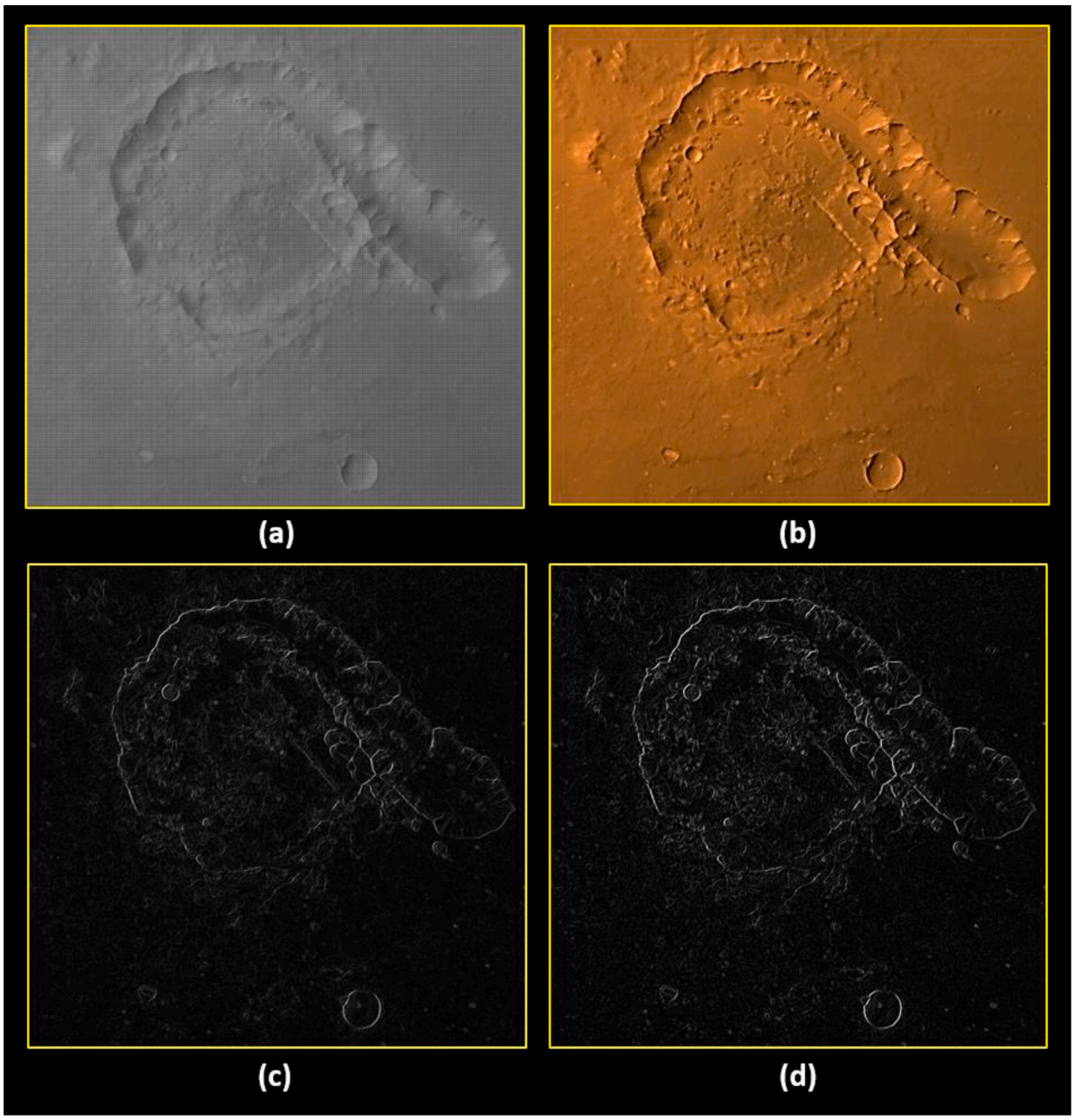

Fig. 9. Direct Feature Extraction at Pital Crater (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

that

$$D \approx D^R \approx D^G \approx D^B \tag{7}$$

It indicates that gradients of multispectral planetary remote sensing images can be computed using any one of the three spectral wavelength channels as long as the CDC assumption holds. So even if two color samples are missing in a pixel location, the gradient of that location in horizontal and vertical directions can be computed directly from the MCC Bayer Pattern raw image using the nearest color spectra values, as follows.

$$D_{x(1,2)} = D^B_{x(1,2)} = I^B_{(1,3)} - I^B_{(1,1)} \tag{8}$$

$$D_{y(1,2)} = D^R_{y(1,2)} = I^R_{(2,2)} - I^R_{(0,2)} \tag{9}$$

The above concept can be extended to other gradient detectors that have alternating zero and non-zero coefficients, and can be used to extract prominent feature targets directly from Bayer Pattern image. Sobel Operator shown in Fig. 1(b) is applied on MCC Bayer Pattern raw data, and the gradient estimate at pixel location is computed as follows:

$$D^S_{x(1,2)} = I^G_{(0,3)} + 2{*}I^B_{(1,3)} + I^G_{(2,3)} - I^G_{(0,1)} - 2{*}I^B_{(1,1)} - I^G_{(2,1)}$$

$$= \left(I^G_{(0,3)} - I^G_{(0,1)}\right) + 2{*}\left(I^B_{(1,3)} - I^B_{(1,1)}\right) + \left(I^G_{(2,3)} - I^G_{(2,1)}\right) \tag{10}$$

$$D^S_{y(1,2)} = I^G_{(2,1)} + 2{*}I^R_{(2,2)} + I^G_{(2,3)} - I^G_{(0,1)} - 2{*}I^R_{(0,2)} - I^G_{(0,3)}$$

$$= \left(I^G_{(2,1)} - I^G_{(0,1)}\right) + 2{*}\left(I^R_{(2,2)} - I^R_{(0,2)}\right) + (I^G_{(2,3)} - I^G_{(0,3)}) \tag{11}$$

where $D^S_{x(1,2)} and D^S_{y(1,2)}$ are gradient values using Sobel Operator at pixel location $(1,2)$ in horizontal and vertical directions respectively, $I^{spectra}_{(x,y)}$ is the intensity value of the spectra (either Red (R), Green (G) or Blue (B)

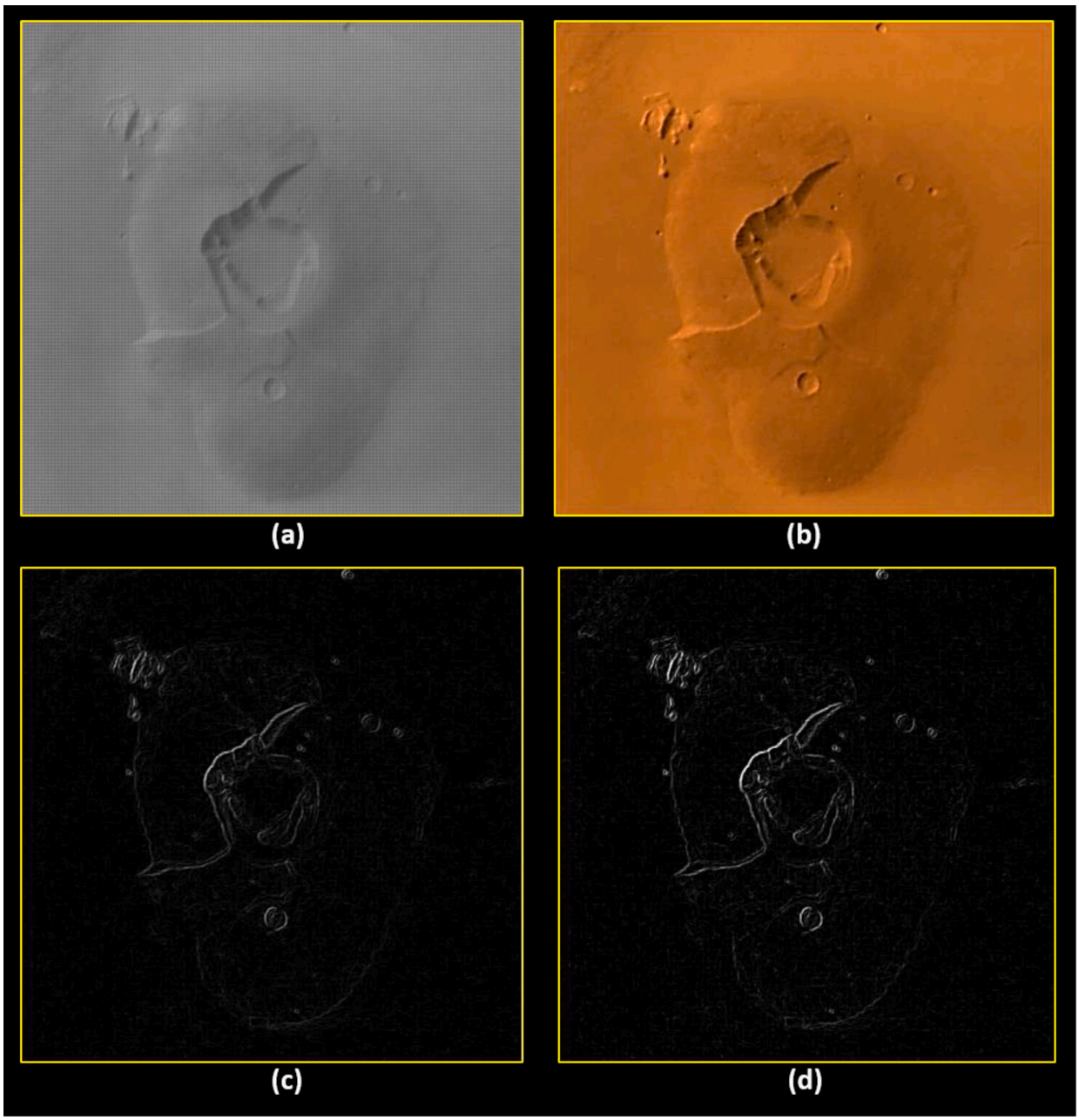

Fig. 10. Direct Feature Extraction at Tharsis Tholus (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

sample) at pixel location (x, y).

The CDC assumption is held by Sobel Edge Operator, and can be used to extract features directly from MCC Bayer Pattern raw image. As per CDC assumption, equation (10) and equation (11) can be re-written as follows.

$$
\begin{aligned}
D^{S}_{x(1,2)} &\approx \left(I^{\hat{R}}_{(0,3)} - I^{\hat{R}}_{(0,1)}\right) + 2^{*}\left(I^{\hat{R}}_{(1,3)} - I^{\hat{R}}_{(1,1)}\right) + \left(I^{\hat{R}}_{(2,3)} - I^{\hat{R}}_{(2,1)}\right) \\
&\approx \left(I^{G}_{(0,3)} - I^{G}_{(0,1)}\right) + 2^{*}\left(I^{\hat{G}}_{(1,3)} - I^{\hat{G}}_{(1,1)}\right) + \left(I^{G}_{(2,3)} - I^{G}_{(2,1)}\right) \\
&\approx \left(I^{\hat{B}}_{(0,3)} - I^{\hat{B}}_{(0,1)}\right) + 2^{*}\left(I^{B}_{(1,3)} - I^{B}_{(1,1)}\right) + \left(I^{\hat{B}}_{(2,3)} - I^{\hat{B}}_{(2,1)}\right) \qquad (12) \\
D^{S}_{y(1,2)} &\approx \left(I^{\hat{R}}_{(2,1)} - I^{\hat{R}}_{(0,1)}\right) + 2^{*}\left(I^{R}_{(2,2)} - I^{R}_{(0,2)}\right) + (I^{\hat{R}}_{(2,3)} - I^{\hat{R}}_{(0,3)}) \\
&\approx \left(I^{G}_{(2,1)} - I^{G}_{(0,1)}\right) + 2^{*}\left(I^{\hat{G}}_{(2,2)} - I^{\hat{G}}_{(0,2)}\right) + (I^{G}_{(2,3)} - I^{G}_{(0,3)}) \\
&\approx \left(I^{\hat{B}}_{(2,1)} - I^{\hat{B}}_{(0,1)}\right) + 2^{*}\left(I^{\hat{B}}_{(2,2)} - I^{\hat{B}}_{(0,2)}\right) + (I^{\hat{B}}_{(2,3)} - I^{\hat{B}}_{(0,3)}) \qquad (13)
\end{aligned}
$$

where $\widehat{R}, \widehat{G}, and \widehat{B}$ are the missing color components at that particular MCC pixel locations.

The pixel index (1, 2) corresponds to the MCC pixel location that is used for gradient map computation. The period of the MCC pixel locations (1, 2) used for feature extraction computation are shown with rounded corner rectangles with yellow-colored dashed boundaries bounding two regions in the Fig. 3 that is used for gradient magnitude extraction. Hence, it is clear that Sobel edge operator can be used to extract features directly from MCC Bayer Pattern raw image.

It is observed that planetary remote sensing images over Mars surface generally contain less texture information, and it is difficult to extract essential features. The flat areas will have negligible difference estimate between two same spectra pixels using Sobel Edge Operator. Contrary to this, the high texture regions such as crater boundary wall and terrain

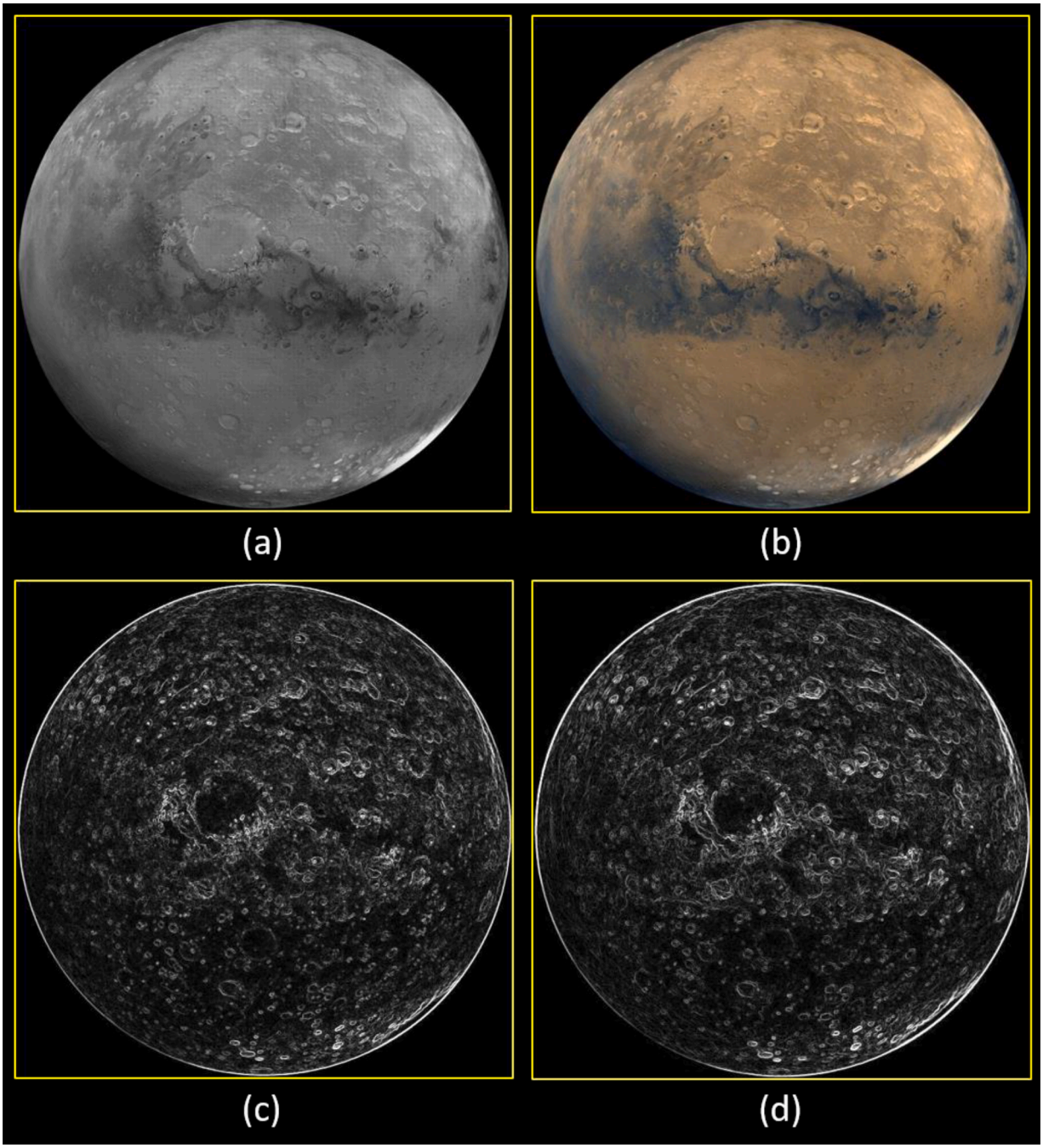

Fig. 11. Direct Feature Extraction of Mars Full Disc Image captured by Viking Orbiter Mission (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

slopes have substantial difference between two same spectra pixels, and will come out as sharp edge information in gradient extracted from MCC Bayer Pattern image. Fig. 4 shows the Gradient Magnitude Map (GMM) extracted from MCC Bayer Pattern raw image and MCC demosaic image using Sobel Edge Operator. It is observed that feature extracted from MCC Bayer data is at par with the gradient feature extracted from MCC demosaic image. At some locations, GMM from Bayer Pattern visually appears to be little faint because of only single spectra sample value availability at each pixel location. Nevertheless, overall GMM from MCC Bayer Pattern provides the required abstraction of image information, which is vital input for object recognition and direct planetary image co-registration. The gradient difference image shown in Fig. 4 (e) demonstrates that there is negligible deviation between GMM map from MCC Bayer Pattern and GMM map from MCC demosaic. The region shown is part of Valles Marineris, which is the largest canyon system of our solar system. The perimeter of areo-morphic structure is traced accurately in GMM from MCC Bayer Pattern raw image, and provides us an insight into the major ramps or slopes around that areo-morphological region. Bayer Pattern Color Filter Array is mostly used for planetary remote sensing sensor and the methodology developed is primarily evaluated in those types of remote sensing images. However, the methodology developed is generalized well to color filter array patterns, such as the RYYB filter and CYYM filter, which follows color difference constancy as also being used in other planetary remote sensing camera systems. So, the technique developed for direct feature extraction and co-registration is scalable.

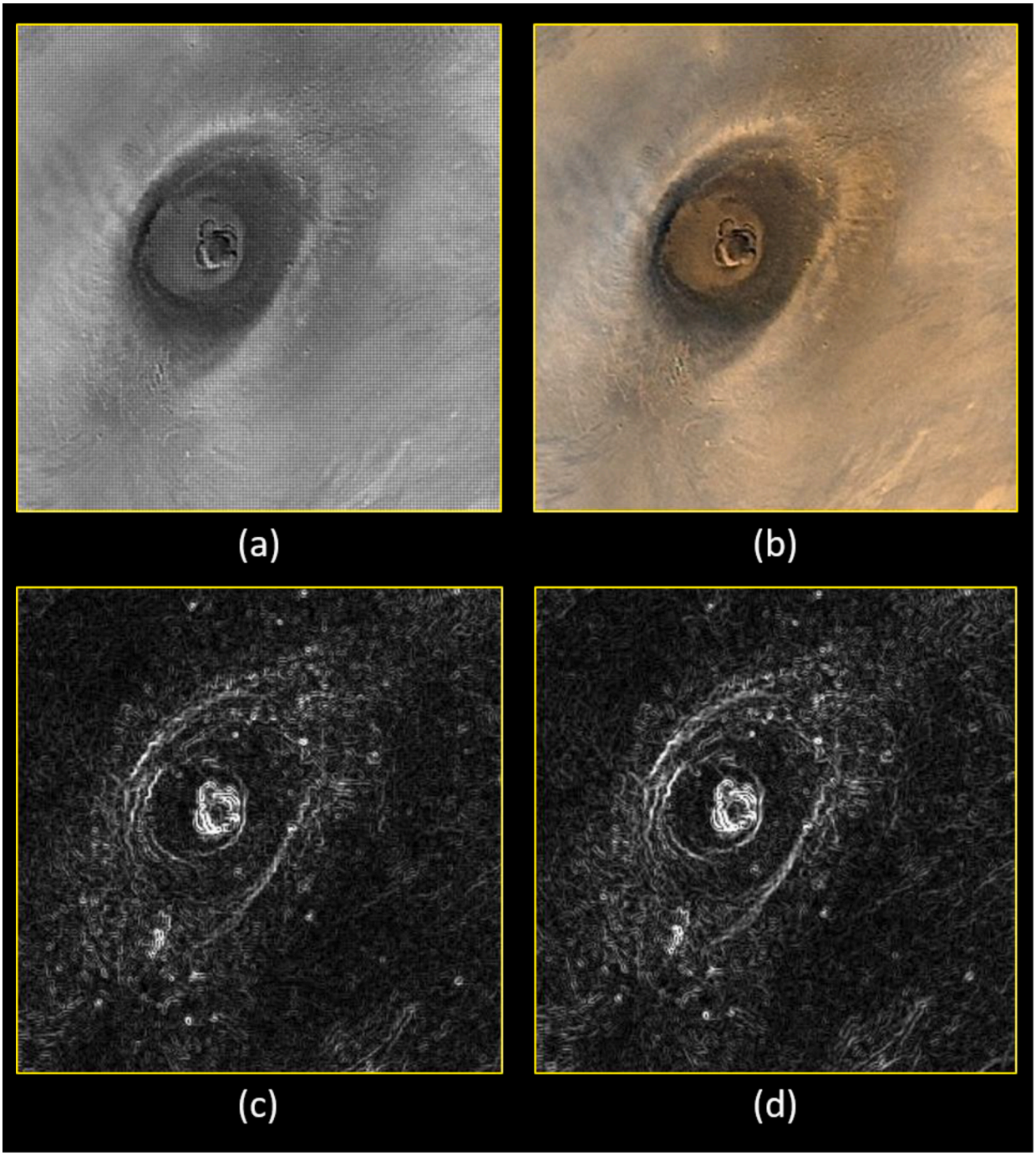

Fig. 12. Direct Feature Extraction of Mars Image captured by Mars Global Surveyor Mission (a) MCC Bayer Pattern Image (b) MCC Demosaic Image (c) GMM from MCC Bayer (d) GMM from MCC Demosaic.

2.2. GI-SIFT feature matching for planetary images

The image co-registration using standard Scale Invariant-Feature Transform (SIFT) initiates with scale space construction using Gaussian Kernel (Goncalves, Corte-Real & Goncalves, 2011; Zhong, Liu & Chen, 2015). If the MCC Bayer Pattern raw image is directly Gaussian filtered, then the mosaic structure of Color Filter Array (CFA) is destroyed, and we may loss essential spectra information which is crucial for feature matching. To overcome this situation, we propose a Mode-Mean based Patch Filler to recover missing color samples directly from CFA without performing any complex interpolation or demosaicking process. Mode is the maximum number of occurrence(s) of a color sample in a patch. If the maximum occurrence for a mode value in a patch is more than a sample score limit, then we fill the missing color samples with the computed mode value. If the sample score is less than a pre-defined limit, then patch mean is computed with available color spectra. The missing values are then filled with that average estimate. In our case, we have selected a patch of 5*5 to compute missing components at each MCC Bayer Pattern pixel location. It is empirically found that the specified patch window contains sufficient single spectra samples for missing color recovery. In the next step, we segregate red, green and blue pixel samples, and compute the mode-mean for the patch under consideration. Empirically, it is found that sample score limit l of three gives optimum result for patch filler. The Bayer intensity image and

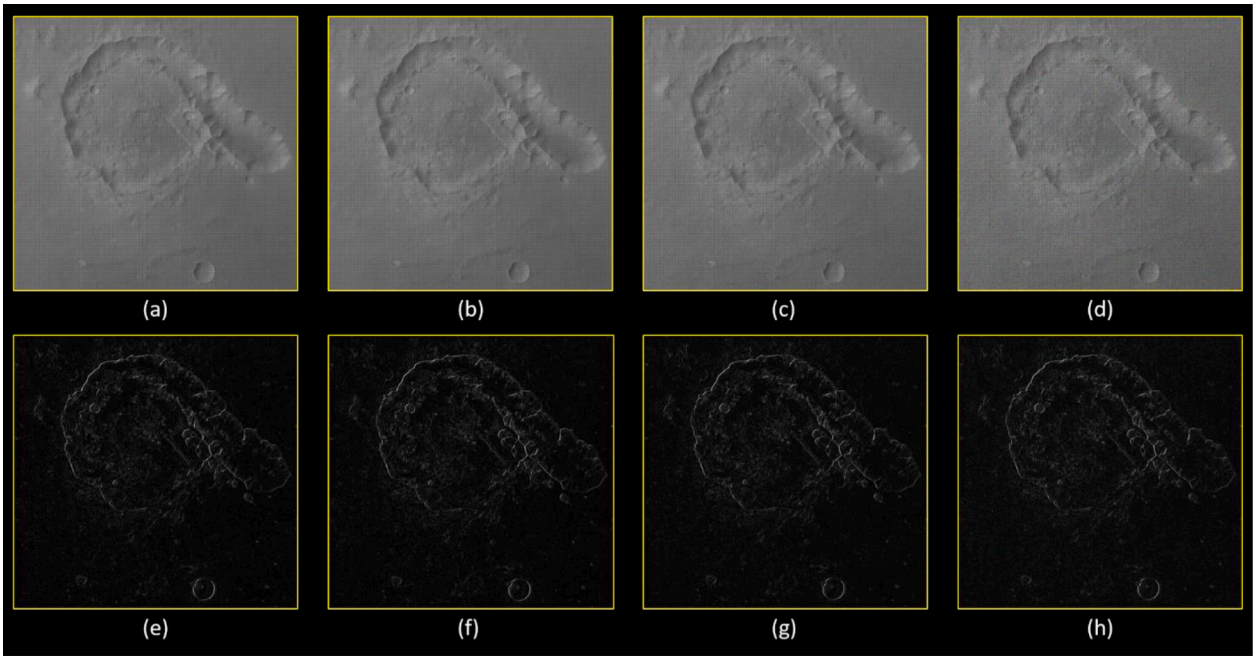

Fig. 13. Feature extraction performance on Pital Crater in presence of varying Gaussian Noise (a) MCC Bayer Pattern Raw Image without noise (b) Raw image with Gaussian Noise σ = 2.0 (c) Raw image with Gaussian Noise σ = 5.0 (d) Raw image with Gaussian Noise σ = 10.0 (e) Feature extracted image without noise (f) Feature extracted image with noise σ = 2.0 (g) Feature extracted image with noise σ = 5.0 (h) Feature extracted image with noise σ = 10.0.

GMM is generated from the filled sample at each MCC image pixel, and can be used directly for feature matching and co-registration of planetary Bayer Pattern images. Missing spectra value S_m in a patch p is computed as follows:

$$(S_m, p) = \{\mu, l \leq 3$$

$$\Omega, l > 3\} \tag{14}$$

where μ = patch mean, Ω = patch mode.

To handle radiometric intensity differences in MCC Bayer Pattern raw images, Gradient Intensity based SIFT (GI-SIFT) feature matching is proposed. GI-SIFT constructs a relation, as given in Equation-15 and Equation-16, between smooth image L and input Bayer intensity raw image I at a pixel location (x, y) using Gaussian kernel across multiple scales. It is able to handle image noise, and also shows good performance in the low Signal-to-Noise Ratio (SNR) imaging conditions.

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \tag{15}$$

$$where G(x, y, \sigma) = \frac{1}{2\pi\sigma} e^{-\left(x^2+y^2\right)/2\sigma^2} \tag{16}$$

The intensity image is Gaussian filtered at scale space level to construct Gaussian pyramid, and gradient magnitude is extracted directly from MCC intensity image using Sobel Edge Detector. Gradient Magnitude of MCC Intensity image D^1_σ is defined as:

$$D^1_\sigma = \sqrt{\left(D^1_{\sigma,x}\right)^2 + \left(D^1_{\sigma,y}\right)^2} \tag{17}$$

where σ is the Gaussian scale space; $D^1_{\sigma,x}$ and $D^1_{\sigma,y}$ are respectively the derivatives of MCC intensity image in horizontal and vertical directions with scale σ.

We now define the gradient orientation O^2_σ and gradient magnitude D^2_σ as follows:

$$O^2_\sigma = \arctan\left(\frac{D^2_{\sigma,y}}{D^2_{\sigma,x}}\right) \tag{18}$$

$$D^2_\sigma = \sqrt{\left(D^2_{\sigma,x}\right)^2 + \left(D^2_{\sigma,y}\right)^2} \tag{19}$$

where $D^2_{\sigma,x}$ and $D^2_{\sigma,y}$ are the derivatives of gradient magnitude of MCC intensity image.

The gradients are computed using Sobel edge operator, which is used for orientation assignment and descriptor extraction for feature matching between MCC Bayer Pattern raw image and Mars image reference. Fig. 5 shows the outputs generated directly from MCC Bayer Pattern raw image, which are inputs for GI-SIFT based feature matching. The MCC Bayer Pattern raw image is used to reconstruct the intensity image from proposed patch filler technique and shows a better representation of MCC Bayer Pattern raw data. The corresponding GMM is generated from MCC Bayer intensity raw image.

2.3. FSS-RANSAC outlier removal technique

GI-SIFT feature matching still suffers from a lot of spurious matched keypoints in planetary images, which induce error in final transformation parameter estimation. Random Sample Consensus (RANSAC) is a powerful outlier removal technique that generates candidate solutions by using the minimum number of observations (control points) required to estimate the underlying model parameters (Misra et al., 2021a; Misra et al., 2022). RANSAC operates in two phases: "Hypothesis Generation" where it randomly picks set of matched keypoints, and "Hypothesis Evaluation" where it evaluates if the chosen hypothesis meets the required inlier candidate whose parameter model error is within a predefined threshold or not. RANSAC algorithm steps are as follow.

1) Randomly select the minimum number of points required to determine the model parameters.
2) Solve for the parameters of the model.
3) Determine how many points from the set of all points fit with a predefined tolerance.

Table 3
Image Quality Metrics Description and Interpretation for Planetary Feature Extraction Assessment.

S. No.	Image Quality Metrics	Description	Interpretation
1.	Gradient Magnitude Similarity Deviation (**GMSD**) (Xue et al., 2013)	GMSD is primarily used for evaluating the sharpness and fidelity of an image by comparing the gradient magnitude (edge information) of the reference image (demosaic image) to the gradient magnitude of a processed image (direct feature Bayer pattern image).	Lower GMSD score indicates higher similarity in gradient magnitude, which generally corresponds to better image quality, as the edges and details in the two images are similar.
2.	Mean Structural Similarity Index Map (**MSSIM**) (Wang et al., 2004)	MSSIM metric is an extension of the Structural Similarity Index (SSIM) and provides a map of SSIM values over an entire image. MSSIM is calculated by computing the SSIM between corresponding local image patches and then averaging these local SSIM values to obtain a single value representing the structural similarity between the two images.	Higher MSSIM score indicates closer structural similarity between two images.
3.	Peak Signal-to-Noise Ratio (**PSNR**) (Welstead, 1999)	PSNR measures the ratio between the maximum possible power of a signal (the peak signal) and the power of corrupting noise that affects the quality of the signal. PSNR is typically calculated using the mean squared error (MSE) between the original image (demosaic image) and the reconstructed image (direct feature extraction Bayer pattern image)	Higher PSNR value implies that the processed image is closer in quality to the original image.
4.	Canberra Distance (**CD**) (Maini & Aggarwal, 2018)	The Canberra Distance (CD) is a metric used to measure the dissimilarity or similarity between two images based on the pixel-wise patches. Each image patch is treated as a set of pixel values, and CD calculates the dissimilarity between corresponding pixels in the two images.	Lower CD value indicates that direct feature extraction from Bayer pattern raw planetary remote sensing image is close to demosaic planetary image.
5.	Sum of Absolute Difference (**SAD**) (Xiao et al., 2023)	SAD or L1 distance is a widely used metric for measuring the dissimilarity or difference between images or image patches. SAD aggregates the absolute differences across all corresponding patch pairs in the images, resulting in a single value that represents the overall dissimilarity between the images.	Lower SAD value indicates greater similarity between images.

4) If the fraction of the number of inliers over the total number of points in the set exceeds a predefined threshold, re-estimate the model parameters using all the identified inliers and terminate.
5) Otherwise, repeat steps 1 through 4 (maximum N times).

The number of iterations, N, is chosen high enough to ensure that at least one of the sets of random samples does not include an outlier. RANSAC uses a portion of data, not whole data. If the selected data are all inliers, they can entail a hypothesis close to the truth. This assumption leads the necessary number of iterations N enough to pick up all inliers' samples at least once as follows.

$$N = \frac{\log\alpha}{\log(1-\gamma^m)} \tag{20}$$

where α = failure probability, m = number of matched keypoints to generate a hypothesis,

γ = probability to pick up an inlier, that is, ratio of inliers to whole sample data (inliers ratio).

However, the inliers ratio γ is unknown in many practical situations, which is necessary to be determined by users. RANSAC converts an estimation problem in the continuous domain into a selection problem in the discrete domain. The problem is now to select the most suitable pair from among a huge number of pairs. In literature, few procedures are present to accelerate the performance of RANSAC (Wu et al., 2014, Zheng et al., 2021). However, the updated RANSAC algorithms do not pick accurate matched pairs free from outliers for planetary remote sensing images. To improve the RANSAC performance to the next level, we propose a novel method FSS-RANSAC that determines the Feature Similarity Score (FSS) of the matched keypoints, and removes the gross outliers beforehand. The Coefficient of Variation (CoV) is determined by setting a local window around the keypoint in both, input MCC Bayer intensity image and reference image, as follows.

$$CoV = \{(\beta/\mu)^*100, w\} \tag{21}$$

where β = standard deviation, μ = mean, w = window size.

$$FSS_p = CoV_{ip} - CoV_{rp} \tag{22}$$

where FSS_p = Feature Similarity Score at matched pixel p, CoV_{ip} = Coefficient of variation at input MCC pixel location ip, CoV_{rp} = Coefficient of variation at reference pixel location rp.

2.4. Direct image Co-Registration workflow

The methodology developed is represented as an image processing pipeline, which takes MCC Bayer Pattern raw image as input and generates MCC intensity raw image using Mode-Mean Combo Patch Filler with the sharp edges being detected using Sobel Edge Operator. The proposed algorithm applied Sobel operator to extract feature directly from Bayer pattern raw planetary image. This is a novel approach to extract essential traits from raw Bayer pattern planetary data, which is not yet explored in planetary science community and can be used for next level of image processing task such as multi-temporal image co-registration. In addition, direct planetary feature extraction can be used for diverse planetary space exploration studies such as crater counting and areomorphological feature analysis. In parallel, MDIM 2.1 Mars reference image is ingested into the image co-registration processing workflow. The reference image is geometrically transformed. Further, the region of interest is cropped and resampled to the spatial resolution of MCC, and the reference GMM is generated. GI-SIFT based feature matching is performed to extract stable matched keypoints. Empirically, it is found that the window size of 5 provides optimum CoV estimate. FSS threshold of 0.8 is set to extract putative keypoint correspondences. The outliers are removed with the proposed FSS-RANSAC technique. The pruned matched points are used to estimate the affine

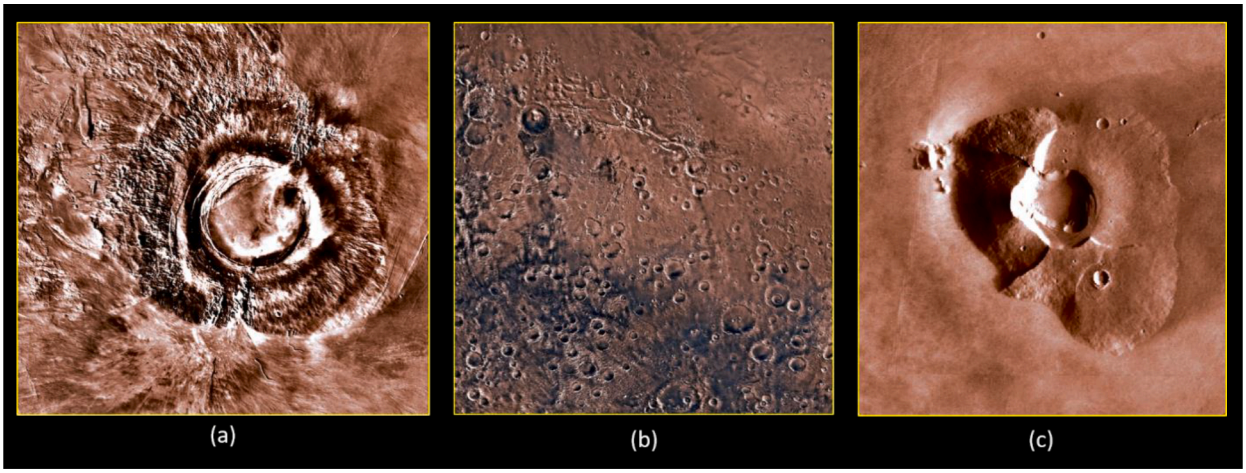

Fig. 14. MDIM 2.1 extracted and transformed ROI reference (a) Arisa Mons (b) Gale Crater (c) Tharsis Tholus.

Table 4
Quantitative Comparison of Gradient Map from MCC Bayer Pattern Raw Image and MCC Demosaic Image.

Gradient Map	Region Covered	GMSD ↓	MSSIM ↑	PSNR ↑	CD ↓	SAD ↓
MCC Bayer Pattern Image	Arisa Mons	0.152	0.842	22.923	0.968	0.803
	Gale Crater	0.138	0.782	27.731	0.941	0.692
	Tharsis Tholus	0.115	0.812	29.324	0.929	0.812
MCC Demosaic Image	Arisa Mons	0.145	0.851	22.952	0.961	0.782
	Gale Crater	0.131	0.786	27.781	0.936	0.678
	Tharsis Tholus	0.102	0.818	29.335	0.925	0.793

Table 5
Feature Extraction Performance at different noise levels.

Pital Crater Feature Extracted Image at different noise level	GMSD ↓	MSSIM ↑	PSNR ↑	CD ↓	SAD ↓
No Gaussian Noise	0.141	0.912	28.923	0.916	0.731
Gaussian Noise $\sigma = 2.0$	0.143	0.902	28.742	0.921	0.739
Gaussian Noise $\sigma = 5.0$	0.145	0.894	28.553	0.928	0.744
Gaussian Noise $\sigma = 10.0$	0.148	0.890	28.422	0.951	0.765

transformation parameters (Misra et al., 2012b), and MCC Bayer Pattern raw image is resampled using cubic convolution kernel such that MCC raw data is spatially aligned with respect to the Mars planetary reference. Fig. 6 shows the direct image co-registration workflow developed for MCC Bayer pattern raw image. The Color Difference Constancy assumption holds true for most of the color filter array pattern images used for remote sensing camera design. So, the technique developed for feature extraction and co-registration will work with accuracy and reliability.

3. Experimental results

The direct feature extraction approach is evaluated at different regions of Mars surface captured by Indian MCC instrument. MCC onboard MOM spacecraft, is capable to capture image surface features of Mars at varying resolutions, and also to monitor dynamic behavior of dust-storms and cloud patterns. MCC can take snapshot of Mars Full Disc at around 4.0 km spatial resolution. Also, MCC can take closer view of a particular areo-morphological structure at better than 15.0 m' spatial resolution. This unique imaging scheme of MCC helps to perform our experiments on different spatial resolution datasets. The MCC Bayer

Table 6
Image Quality Metrics Description and Interpretation to assess Planetary Image Co-Registration Performance.

S. No.	Metrics	Description	Interpretation
1.	Root Mean Square Error (**RMSE**) (Chai & Draxler, 2014)	RMSE is a widely used metric for evaluating the quality of image co-registration. RMSE is used to measure the accuracy of this alignment by comparing the pixel-wise differences between the reference image and the co-registered image.	Lower RMSE indicates better alignment or co-registration, as it means that the co-registered image is closer to the reference image.
2.	Correct Matching Ratio (**CMR**) (Zhang, 2020)	CMR is a metric used in image co-registration to evaluate the accuracy of the matching or alignment of two images. CMR is used to assess how well the matching process has been performed by calculating the ratio of correctly matched points to the total number of matched points.	Higher CMR indicates a higher level of accuracy in the co-registration process, as it means that a larger proportion of the matched points are correctly aligned.

Table 7
Image Co-Registration RMSE and CMR Accuracy Comparison.

Image Data Used	Region Covered	RMSE (in pixel)↓	CMR↑
	Arisa Mons	0.452	0.923
	Gale Crater	0.338	0.956
	Tharsis Tholus	0.515	0.911
	Arisa Mons	0.312	0.928
	Gale Crater	0.268	0.961
	Tharsis Tholus	0.435	0.931

Table 8
Image Co-Registration performance of Arisa Mons by running the proposed method multiple times.

Execution Number	RMSE (in pixel)↓
Run-1	0.452
Run-2	0.456
Run-3	0.451
Run-4	0.454

Table 9
Multi-temporal Image Co-Registration Performance at complex background and varying illumination conditions.

Region Covered	RMSE (in pixel)↓
Arisa Mons Multi-temporal Image-1	0.452
Arisa Mons Multi-temporal Image-2	0.406
Arisa Mons Multi-temporal Image-3	0.482
Gale Crater Multi-temporal Image-1	0.338
Gale Crater Multi-temporal Image-2	0.381
Gale Crater Multi-temporal Image-3	0.318

Pattern raw data details used for our experiment are given in Table 1. The MCC raw data is packed as per Planetary Data System (PDS) standard, and is freely available to download through MOM LTA web browse portal (MOM, 2022) for scientific user community. Mars surface contains different types of areo-morphic structures such as Mons, Craters, Canyon system, Fractures and Grabens. The datasets of diverse types of Mars surface features are taken for our direct feature extraction exercise so that we can generalize the performance of the proposed method. In addition, the datasets are selected at different spatial resolution to get more insight on feature extraction performance at different scales. The experimental parameters settings of the proposed method are specified in Table 2 to evaluate the feature extraction performance. The experimental setting is done in such a way that the direct feature extraction method provides optimal results at different regions, scales and imaging conditions. The image co-registration performance is assessed in varying resolution MCC Bayer Pattern raw images and is compared with it's demosaic counterpart.

3.1. GMM visual quality comparison

The gradient information is extracted from MCC Bayer Pattern images and generated GMM directly from raw data. MCC Bayer raw image is demosaicked to generate full RGB color image, and GMM is extracted from complete color spectrum as well. Fig. 7, Fig. 8, Fig. 9 and Fig. 10 shows GMM from MCC Bayer Pattern raw image and GMM from MCC

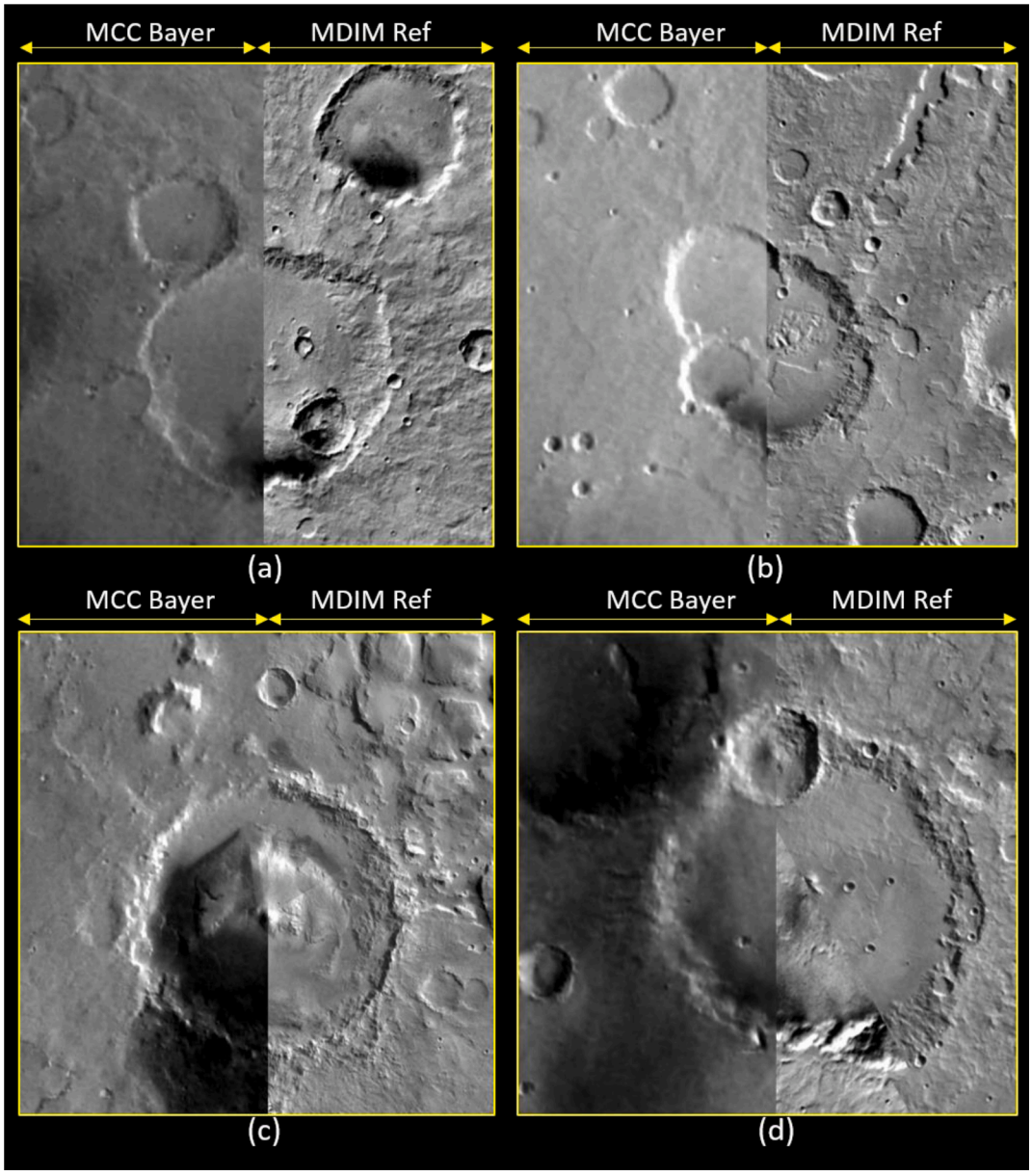

Fig. 15. Visual Assessment of Sub-Pixel Alignment between MCC Bayer and MDIM Reference at different types of crater over planet Mars (a) Crater-01 (b) Crater-02 (c) Crater-03 (d) Crater-04.

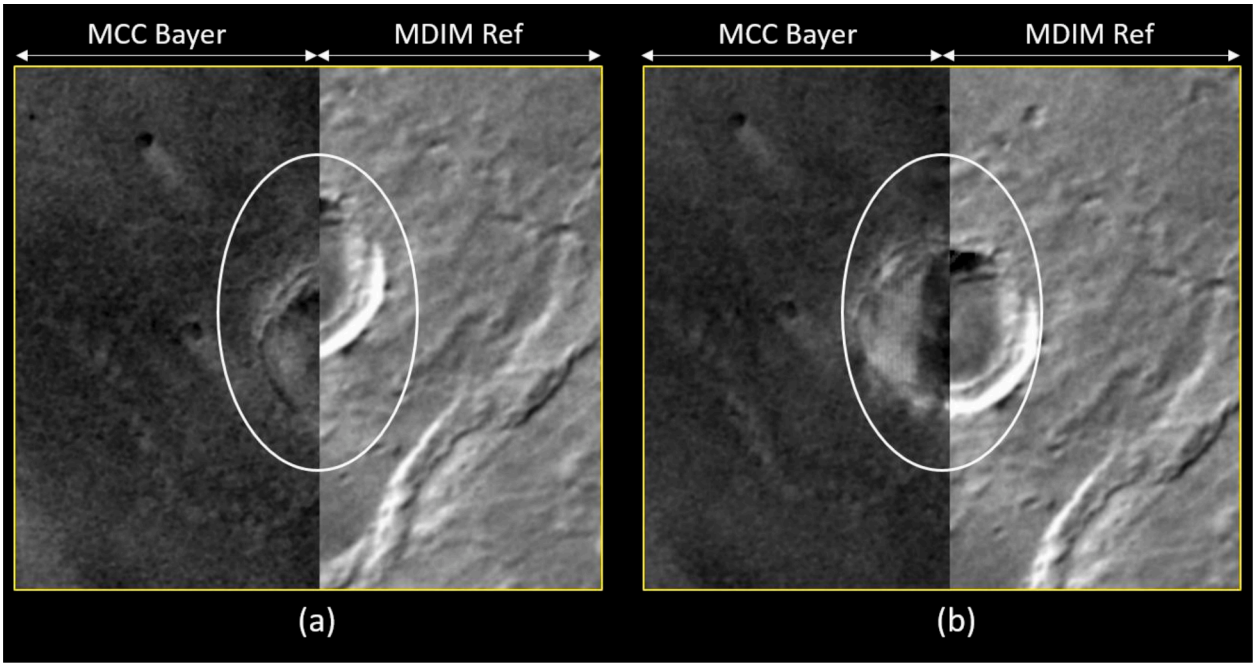

Fig. 16. Planetary Image Co-Registration Performance over Martian Crater (a) Before Registration (b) After Registration.

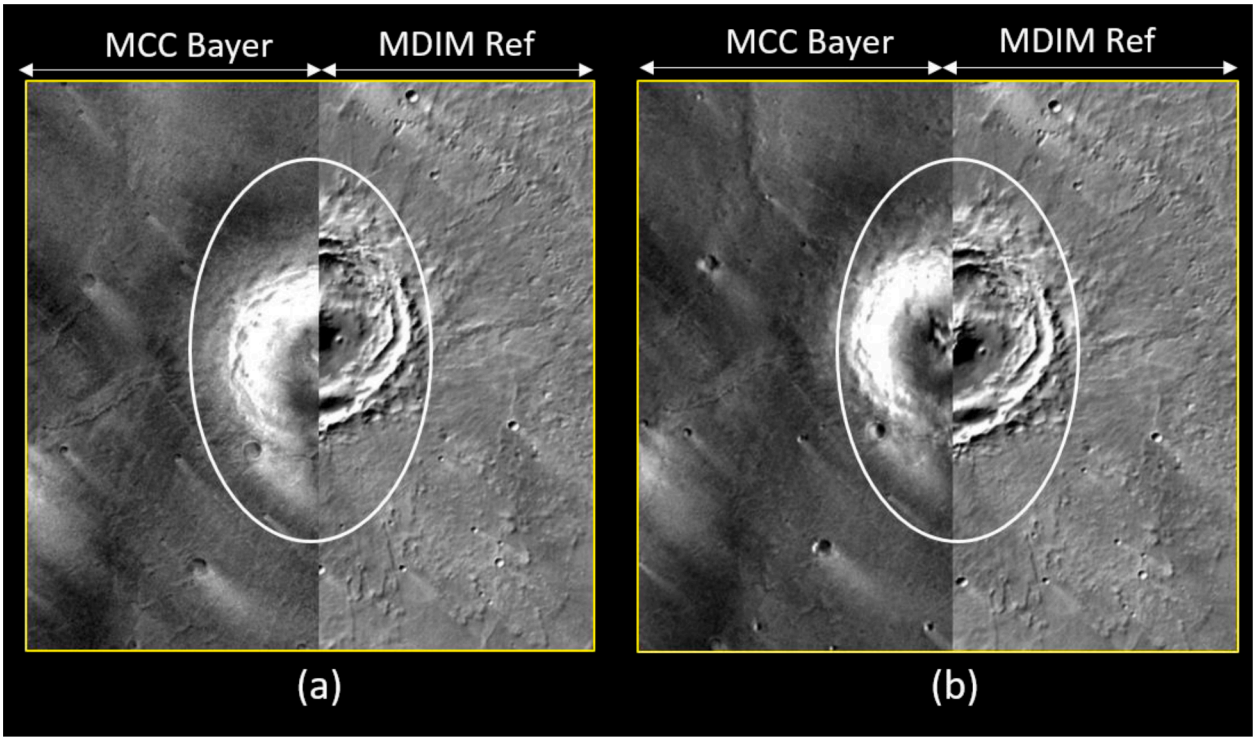

Fig. 17. Planetary Image Co-Registration Performance over Kinkora Crater (a) Before Registration (b) After Registration.

demosaic image for visual assessment. Different areo-morphological structures are considered for our experiment, which are at different spatial resolution covering physically far apart locations on Mars. In Fig. 7, the outer ring of Arisa Mons is extracted directly from MCC Bayer raw image, and is at par with GMM generated from MCC demosaic image. In Fig. 8, the circumference of Gale Crater is tracked accurately in GMM from MCC Bayer data, which was the landing site of NASA's Curiosity rover. Pital Crater shown in Fig. 9 is an impact crater imaged by MCC at relatively higher spatial resolution. The wall of the Pital crater in GMM is exactly extracted from MCC Bayer image, and is found to have a diameter of around 40 km. Tharsis Tholus is a shield volcano shown in Fig. 10, which looks like a doom shaped areo-morphic structure. The direct GMM shows the territory covered by the volcano over Mars surface. The experimental visual results show that proposed method extract feature unambiguously from MCC Bayer pattern raw planetary images. Crater, Mons and Valles are prominent areomorphological features on the Martian planetary surface. It is found that methodology developed extract the feature precisely that is independent of different

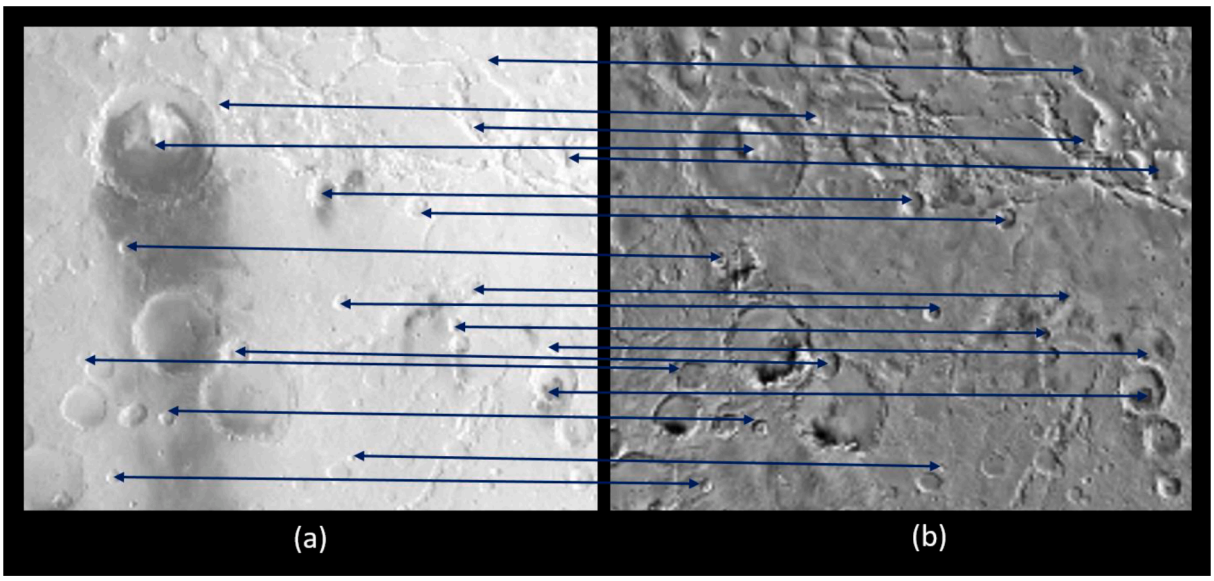

Fig. 18. FSS-RANSAC Outlier Removal Performance around Gale Crater (a) MCC Bayer Intensity Image (b) MDIM 2.1 Reference Image.

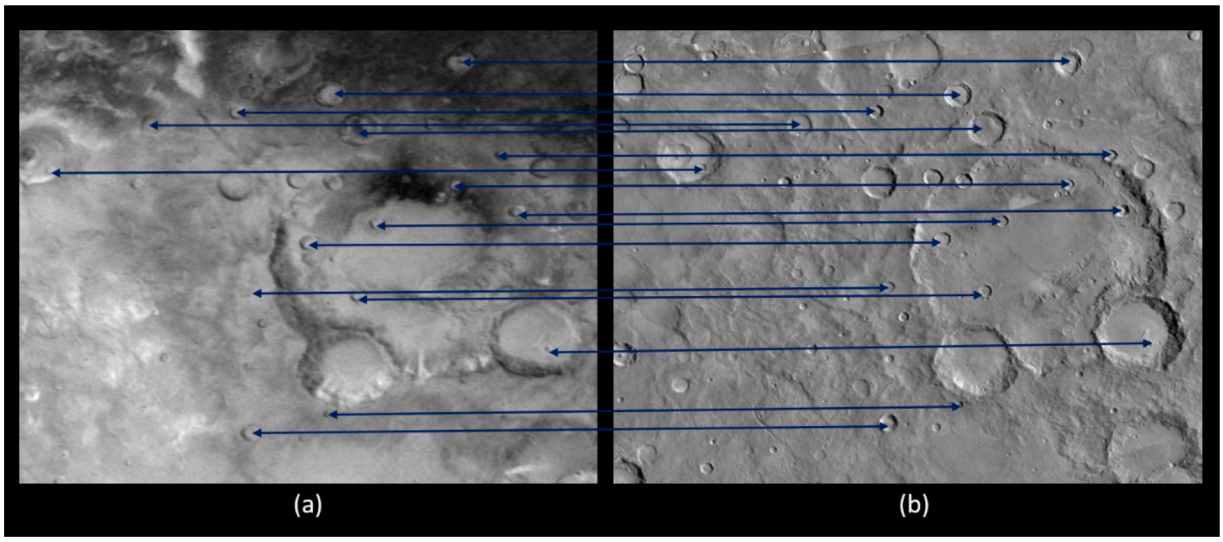

Fig. 19. FSS-RANSAC Outlier Removal Performance around Martian Craters (a) MCC Bayer Intensity Image (b) MDIM 2.1 Reference Image.

topographical feature targets. In addition, the gradient magnitude feature map from raw Bayer planetary image shows very close similarity to the feature extracted from demosaic color planetary image.

The proposed method is applied on different types of remote sensing images. In the experimental section, we have introduced planetary remote images from Viking Orbiter Mission (Snyder, 1977) and Mars Global Surveyor Mission (Albee, Palluconi, & Arvidson, 1998). The simulated Bayer pattern image is generated from color multi-spectral images for evaluation. Fig. 11 shows Mars full disc Bayer pattern image and color multispectral image captured by Viking orbiter. Fig. 11 (c) and Fig. 11 (d) represents the corresponding gradient magnitude map generated from Bayer pattern image and color multispectral image. The same procedure is tested with Mars Global surveyor image and GMM is generated from Bayer pattern planetary image as well as from color multispectral image shown in Fig. 12. It is found that feature extraction from Bayer pattern images shows close resemblance with respect to color multispectral images from Viking orbiter and Mars Global surveyor images. In addition, the performance of the proposed method is evaluated under different noise conditions. For simulation, different level of Gaussian noise is added in the MCC raw Bayer pattern images. The feature is directly extracted from Bayer pattern noisy planetary images. The image shown is of Pital crater on Mars surface. It is visually observed from Fig. 13 that features are extracted properly from MCC Bayer pattern images even in the presence of Gaussian noise.

3.2. *Quantitative assessment and Co-Registration performance*

The quantitative evaluation of GMM is performed using different image quality metrics. Gradient Magnitude Similarity Deviation (GMSD) (Xue et al., 2013) is a popular metric to assess the similarity of gradient magnitudes and can indicate how much deviation the direct gradient map from MCC Bayer pattern raw image with respect to the reference gradient map. The smaller the GMSD value closer the gradient maps compared. GMSD metric value close to 0 indicates perfect agreement (Mason et al., 2019) which is difficult to achieve in gradients between multi-temporal planetary remote sensing images where dynamic surface phenomenon also plays a critical role. Mean Structural Similarity Index Map (MSSIM) (Wang et al., 2004) is a standard full reference image

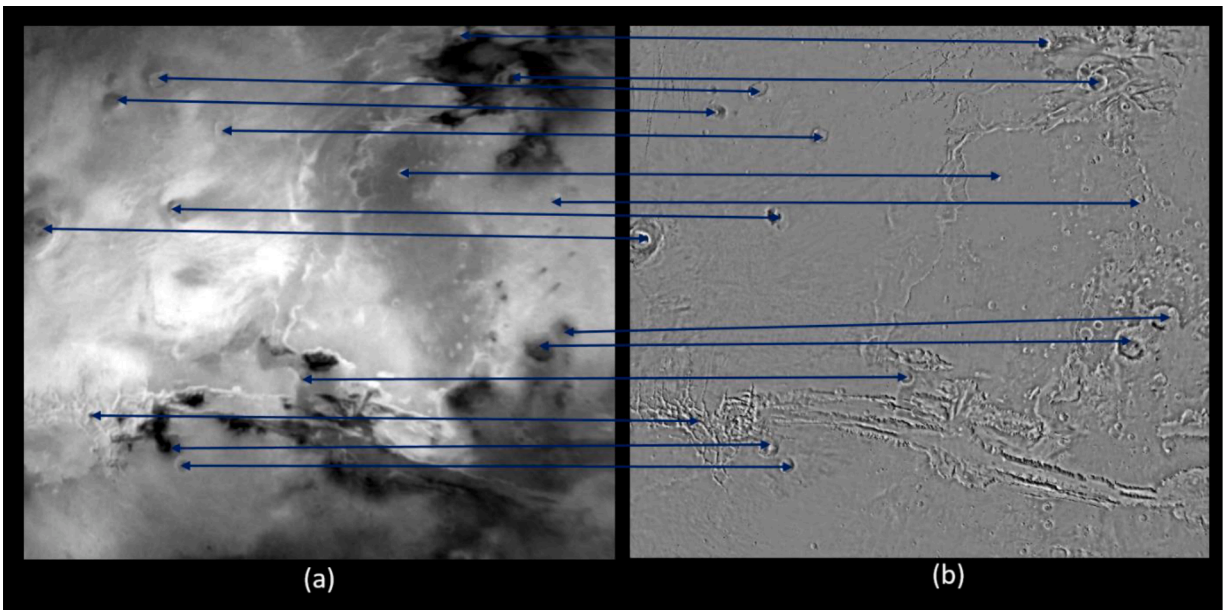

Fig. 20. FSS-RANSAC Outlier Removal Performance around Valles Marineris (a) MCC Bayer Intensity Image (b) MDIM 2.1 Reference Image.

Table 10
RMSE Comparison using different Image Registration Techniques over Bayer Pattern Raw Planetary Images.

S.No.	Mars Region Covered	Harris + RANSAC	SIFT + RANSAC	ORB + MSC	SURF + MLESAC	KAZE+ MAGSAC	Semantic DNN	MU-NET	Proposed Method
1.	Arisa Mons	0.786	0.576	0.662	0.584	0.612	0.598	0.526	0.452
2.	Gale Crater	0.581	0.674	0.563	0.498	0.553	0.488	0.491	0.338
3.	Tharsis Tholus	0.855	0.611	0.681	0.675	0.674	0.591	0.575	0.515

Table 11
CMR Comparison using different Image Registration Techniques over Bayer Pattern Raw Planetary Images.

S.No.	Mars Region Covered	Harris + RANSAC	SIFT + RANSAC	ORB + MSC	SURF + MLESAC	KAZE+ MAGSAC	Semantic DNN	MU-NET	Proposed Method
1.	Arisa Mons	0.865	0.901	0.893	0.889	0.895	0.902	0.891	0.923
2.	Gale Crater	0.898	0.912	0.906	0.934	0.913	0.899	0.925	0.956
3.	Tharsis Tholus	0.861	0.862	0.856	0.857	0.862	0.882	0.857	0.911

Table 12
NCM Comparison using different Image Registration Techniques over Multi-Temporal Bayer Pattern Raw Planetary Images.

S. No.	Image Co-Registration Techniques	Number of Correct Matches (NCM)
1.	Harris + RANSAC	50
2.	SIFT + RANSAC	40
3.	ORB + MSC	85
4.	SURF + MLESAC	80
5.	KAZE + MAGSAC	45
6.	Semantic DNN	65
7.	MU-NET	85
8.	Proposed Method	120

quality metric that compares luminance, contrast and structure among two images. MSSIM can also be used to compare GMM. Peak Signal to Noise Ratio (PSNR) is another stable reference quality metric that is inversely proportional to Mean Square Error (MSE) and direct gradient image is compared with ideal gradient image with maximum possible power (Welstead, 1999). In addition, the image quality metrics such as Canberra Distance (CD) and Sum of Absolute Distance (SAD) show the gradient map performance of Bayer and Demosaic images with respect to reference data (Maini & Aggarwal, 2018; Xiao et al., 2023). Table 3 shows the image quality metrics with their description and interpretation, which are used for assessing the direct feature extraction performance. Fig. 14 shows Mars Digital Image Model (MDIM) 2.1 extracted Region of Interest (ROI) reference of three identified mar terrains used for quantitative assessment. MDIM 2.1 is a global base map of Mars at a spatial resolution of 231 m. MCC images used for the experiment ranges from 292 m to 453 m per pixel covering Arisa Mons, Gale Crater and Pital crater. In general for image registration, reference image selected should have better spatial resolution so that input images can be aligned accurately with respect to the reference. The ROI is cropped from MDIM 2.1 and geometrically transformed to MCC image resolution using standard cubic resampling kernel to generate a smooth corresponding reference image. The reference gradient image from MDIM 2.1 transformed image is generated using Sobel edge detector that act as a reference gradient map to assess GMM from MCC Bayer Pattern raw image and GMM from MCC demosaic image. Table 4 shows the quantitative comparison between extracted GMM from MCC Bayer pattern image and MCC Demosaic image at three different regions of Mars that includes Arisa Mons, Gale Crater and Tharsis Tholus. At different Mars terrain, GMSD, MSSIM, PSNR, CD and SAD estimates are computed and it is observed that metric values of GMM from MCC Bayer Pattern image have close agreement with respect to GMM from MCC Demosaic image.

The image quality metrics such as GMSD, MSSIM, PSNR, CD and SAD

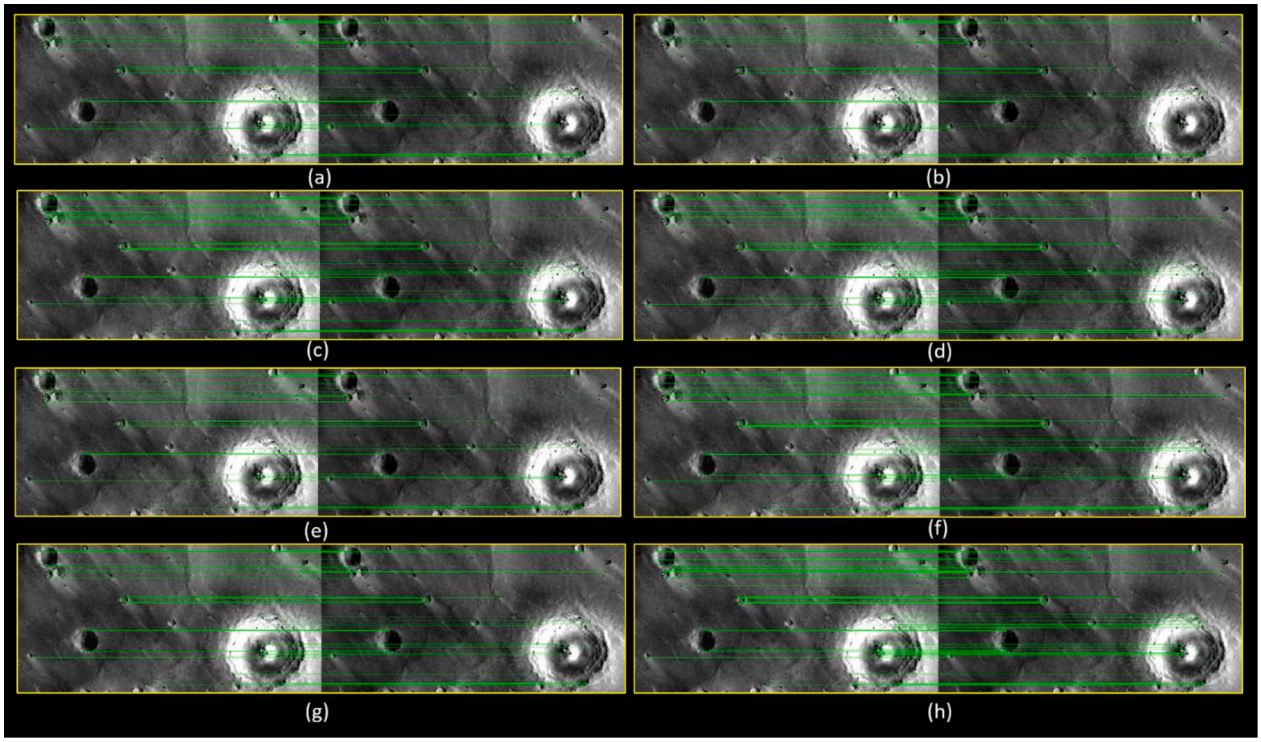

Fig. 21. Feature Matching Performance between multi-temporal planetary images (a) Harris + RANSAC (b) SIFT + RANSAC (c) ORB + MSC (d) SURF + MLESAC (e) KAZE + MAGSAC (f) Semantic DNN (g) MU-NET (h) Proposed Method.

are computed for different Gaussian noise levels to quantify the performance of the methodology in the presence of noise. Table 5 shows feature extraction performance at different Gaussian noise levels. The feature extracted from MCC demosaic image act as a reference for quality metrics computation of Gaussian noise levels. It is found that Pital crater feature are extracted unambiguously at varying noise levels with very less performance degradation.

The MCC Bayer Pattern images falling over Arisa Mons, Gale Crater and Tharsis Tholus are co-registered with MDIM 2.1 transformed ROI shown in Fig. 14, using proposed GI-SIFT feature matching. The corresponding MCC demosaic images over these Martian sites are also matched and co-registered with GI-SIFT technique. The MCC demosaic image is generated through end-to-end radiometric correction, which, along with demosaicking method, also includes PRNU correction, deblurring, denoising, and color balancing procedures. Correct Matched Points are determined in both MCC Bayer Image and MCC Demosaic Image after applying FSS-RANSAC outlier removal method. Root Mean Square Error (RMSE) metric (Chai & Draxler, 2014) is computed using co-registered matched keypoints' correspondence established between MCC image and MDIM 2.1 reference using GI-SIFT and FSS-RANSAC technique. RMSE is defined as follows:

$$RMSE = \sqrt{\left(\frac{1}{N}\sum_{i=1}^{N}\|X_i - \widehat{X}_i\|^2\right)} \tag{23}$$

where N represents number of corrected matched points, X_i is the (x_i, y_i) estimated coordinates in the MCC image, $\widehat{X}_i$ is the $(\widehat{x}_i, \widehat{y}_i)$ actual coordinates in Mars reference image.

Correct Matching Ratio (CMR) is another effective measure for quantifying the image registration output (Zhang, 2020). CMR is defined as follows.

$$CMR = \frac{N_c}{N_T} \tag{24}$$

Here, N_T is the number of matched correspondences after outlier removal, and N_c is the number of exact correct matches obtained by finally removing the false ones. Table 6 shows the image quality metrics description and interpretation that are used to assess the quality of planetary image co-registration performance.

Table 7 shows the RMSE values achieved at different Martian terrains using MCC Bayer Pattern raw image and MCC demosaic image. The image co-registration accuracy achieved with direct MCC Bayer Pattern raw image is found to be very close to that with MCC demosaic image. It is clearly observed that average RMSE achieved using MCC Bayer Pattern raw image is within 0.5 pixel at far apart Martian regions. Table 7 also shows CMR accuracy comparison between MCC Bayer pattern image and MCC demosaic image. It is found that CMR metric computed from MCC Bayer pattern image using the proposed method shows close resemblances with the MCC demosaic image and indicates similar feature matching performance at diverse Martian landscape.

To check the robustness of the proposed method to extract feature directly from raw Bayer pattern planetary image and co-registration of morphological structure, the processing workflow is executed multiple times. Table 8 shows RMSE performance of image co-registration of the Arisa Mons feature on Mars surface. It is found from the Table 8 that even proposed method selects the points randomly, the co-registration performance is consistent enough at the multiple running instances. The multi-temporal imaging acquisitions over Arisa Mons and Gale Crater are also taken into consideration, which is imaged at different lightning conditions due to highly elliptical orbit of Mars Orbiter Mission spacecraft. Table 9 shows the RMSE performance of multi-temporal imaging acquisitions with complex background and insignificant local differences. It is found from the Table 9 that image co-registration performance with the proposed method is similar for multi-temporal images captured at different solar illumination conditions.

Fig. 15 shows horizontal image swipes to assess the registration performance between MCC Bayer Pattern intensity image (left side) and MDIM 2.1 reference map (right side) over diverse types of planet Mars crater. The visual assessment confirms that gradient extracted from MCC

Bayer Pattern intensity image is robust enough to co-register different feature targets over planet Mars at sub-pixel level co-registration accuracy with respect to MDIM 2.1 reference.

The proposed method is evaluated with MCC Bayer and MDIM reference by swiping images over each other at different Martian feature terrain. Fig. 16 (a) shows MCC Bayer pattern raw image before registration is swiped with respect to MDIM reference to visualize the misalignment at the crater region. After planetary image co-registration using the methodology developed, the MCC Bayer image is superimposed over MDIM reference shown in Fig. 16 (b). The methodology is also evaluated at Kinkora Crater, which is a relatively bigger crater. Fig. 17 shows image co-registration performance over Kinkora Crater. It is clearly observed that after registration MCC Bayer image is aligned at sub-pixel level with respect to MDIM reference image.

Fig. 18, 19 and 20 shows FSS-RANSAC performance over Martian terrain. The pruned matched points generated by FSS-RANSAC outlier removal algorithm is consistent at different regions of Martian landscape. Dark blue color lines in Fig. 18 to Fig. 20 show putative keypoint correspondences detected by FSS-RANSAC for final model parameter estimation.

The co-registration accuracy of less than 0.5 pixel is set as an expected goal for planetary remote sensing images. The proposed method is able to extract features effectively to align images at sub-pixel level and achieve the expected target. The co-registration performance is evaluated at different feature targets at Mars terrain and it is found that average RMSE metrics is 0.435 pixel that is within the goal limit set.

3.3. Comparison with State-of-the-Art feature based techniques

The quantitative comparison of the proposed image registration method with other state-of-the-art feature based image registration techniques are carried out at different regions of Mars. The state-of-the-art image registration techniques are better to compare quantitatively as other registration methods also achieve sub-pixel registration accuracy and visually it looks similar. In quantitative analysis, we can get insight that how proposed method registration accuracy is better than other eminent registration techniques at different areomorphological structures of Mars surface. Planetary data pre-processing step extract the features from MCC raw Bayer pattern image and make abstract information ready for Bayer pattern data co-registration. The compared methods adopt the same data pre-processing methodology for comparison. The proposed method is compared with Harris with RANSAC (Misra et al., 2012), SIFT with RANSAC (Hernández & Ortiz, 2020; Misra, Rohil, Moorthi, & Dhar, 2021b), ORB with MSC (Misra et al., 2012b), SURF with MLESAC (Misra et al., 2022) and KAZE with MAGSAC (Alcantarilla, Bartoli & Davison, 2012; Misra et al., 2023). In addition, deep learning based image registration technique is also considered for comparison and evaluation. Deep Neural Network (DNN) with Semantic matching (Li et al., 2021) and multi-scale framework with unsupervised learning method known as MU-NET (Ye et al., 2022) used for planetary image registration task. RMSE (Chai & Draxler, 2014) and CMR (Zhang, 2020) are computed for the image registration methods.

It is found that proposed method is superior in terms of RMSE and CMR than other popular and recent feature based image co-registration techniques. RMSE is found minimum by proposed method in all three Mars regions identified for image registration evaluation and shown in Table 10. In addition, CMR is highest in the proposed method of image registration for Bayer pattern raw planetary images captured over Mars surface and shown in Table 11.

To compare the performance of planetary image co-registration feature matching performance, the Number of Correct Matches (NCM) is computed. NCM between multi-temporal MCC images that cover craters of different dimensions is calculated using different image co-registration methods. The size of the planetary image is 1817 pixels (width) and 886 pixels (height). Table 12 shows NCM values and it is clearly found that proposed method is able to generate highest number of correct feature matching points between multi-temporal planetary images. Fig. 21 shows the feature matching performance visually by different eminent image co-registration techniques. It indicates the distribution of matched feature points by image co-registration approaches. It is found that the proposed method successfully generate the best feature matching performance among all the state-of-the-art image co-registration techniques.

4. Conclusion

The MCC Bayer Pattern raw images are used for direct information extraction of different areo-morphological feature targets over planet Mars. Color Difference Constancy (CDC) assumption leads to generate Gradient Magnitude Map (GMM) that indicates crater boundaries and mons circumference directly from MCC raw data. The Central Difference Gradient Operator and Sobel Edge Operator that follow CDC are applied over MCC raw images covering different Martian terrains. The visual comparison shows that GMM from MCC Bayer Pattern raw image and GMM from MCC demosaic images have negligible deviation. The proposed method is also evaluated on planetary remote sensing images from Viking Orbiter and Mars Global Surveyor, which shows similar feature extraction performance. The quantitative metrics are computed for GMMs of both the versions, and it is found that the direct feature extraction method developed is robust enough to extract the significant information. GI-SIFT feature matching with FSS-RANSAC outlier removal technique is able to co-register MCC Bayer pattern raw image with respect to standard Mars reference at sub-pixel level spatial alignment. The visual quality assessment and quantitative evaluation showcase the planetary image co-registration performance achieved using the proposed methodology. The limitation of the method is that it is able to extract features directly from those types of planetary images which are in the form of Bayer pattern color scheme. The proposed method is suitable for planetary remote sensing images taken at different lightning condition. The technique is evaluated with multi-temporal MCC images that are captured at varying solar illumination conditions due to highly elliptical orbit of Mars Orbiter Mission (MOM) spacecraft. It is found that proposed method extract features seamlessly from MCC raw Bayer pattern image at varying imaging conditions. The direct feature extraction and image co-registration workflow developed for MCC raw image provides the essential traits, which can be used for crater counting, identifying morphological structure area, change detection and perform the subsequent level of image processing operations without much hindrance.

CRediT authorship contribution statement

Indranil Misra: Conceptualization, Methodology, Software, Writing – original draft. **Mukesh Kumar Rohil:** Conceptualization, Formal analysis, Writing – review & editing. **S Manthira Moorthi:** Conceptualization, Formal analysis, Writing – review & editing. **Debajyoti Dhar:** Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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